

REMARK In the above formulae it should be noted that the angle on the RHS is half of the angle on LHS.

$$\therefore \sin 60^\circ = 2 \sin 30^\circ \cos 30^\circ, \cos 120^\circ = \cos^2 60^\circ - \sin^2 60^\circ, \tan 60^\circ = \frac{2 \tan 30^\circ}{1 - \tan^2 30^\circ} \text{ etc.}$$

9.3 TRIGONOMETRIC RATIOS OF THE ANGLE A IN TERMS OF THAT OF ANGLE $\frac{A}{2}$

The relations in section 9.2 are true for all values of the angle A for which the two sides are meaningful. Replacing A by $A/2$ in the above relations, we obtain the following relations:

$$(i) \sin A = 2 \sin \left(\frac{A}{2} \right) \cos \left(\frac{A}{2} \right) \quad (ii) \cos A = \cos^2 \left(\frac{A}{2} \right) - \sin^2 \left(\frac{A}{2} \right)$$

$$(iii) \cos A = 2 \cos^2 \left(\frac{A}{2} \right) - 1 \quad \text{or,} \quad 1 + \cos A = 2 \cos^2 \left(\frac{A}{2} \right)$$

$$(iv) \cos A = 1 - 2 \sin^2 \left(\frac{A}{2} \right) \quad \text{or,} \quad 1 - \cos A = 2 \sin^2 \left(\frac{A}{2} \right)$$

$$(v) \tan A = \frac{2 \tan \left(\frac{A}{2} \right)}{1 - \tan^2 \left(\frac{A}{2} \right)} \quad (vi) \sin A = \frac{2 \tan \left(\frac{A}{2} \right)}{1 + \tan^2 \left(\frac{A}{2} \right)}$$

$$(vii) \cos A = \frac{1 - \tan^2 \left(\frac{A}{2} \right)}{1 + \tan^2 \left(\frac{A}{2} \right)}$$

9.4 TRIGONOMETRIC RATIOS OF THE ANGLE $A/2$ IN TERMS OF $\cos A$

We have,

$$\cos A = 2 \cos^2 \frac{A}{2} - 1 \Rightarrow 2 \cos^2 \frac{A}{2} = 1 + \cos A \Rightarrow \cos \frac{A}{2} = \pm \sqrt{\frac{1 + \cos A}{2}}$$

The sign on RHS depends upon the quadrant in which angle $\frac{A}{2}$ lies.

Also,

$$\cos A = 1 - 2 \sin^2 \frac{A}{2} \Rightarrow 2 \sin^2 \frac{A}{2} = 1 - \cos A \Rightarrow \sin \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{2}}$$

The sign on RHS depends upon the quadrant in which angle $\frac{A}{2}$ lies.

$$\text{Now, } \tan \frac{A}{2} = \frac{\sin \frac{A}{2}}{\cos \frac{A}{2}} = \pm \frac{\sqrt{\frac{1 - \cos A}{2}}}{\sqrt{\frac{1 + \cos A}{2}}} = \pm \sqrt{\frac{1 - \cos A}{1 + \cos A}}$$

The sign on RHS depends upon the quadrant in which angle $\frac{A}{2}$ lies.

REMARK These relations are very useful to find the trigonometric ratios of the angles

$22\frac{1}{2}^\circ, 7\frac{1}{2}^\circ, 11\frac{1}{2}^\circ$ etc.

ILLUSTRATIVE EXAMPLES**LEVEL-1**

Type I ON FINDING THE VALUES OF $\sin 2A$, $\cos 2A$, $\tan 2A$ ETC WHEN VALUES OF $\sin A$ OR $\cos A$ OR $\tan A$ ARE GIVEN

EXAMPLE 1 If $\sin A = \frac{3}{5}$, where $0^\circ < A < 90^\circ$, find the values of $\sin 2A$, $\cos 2A$, $\tan 2A$ and $\sin 4A$.

SOLUTION We have, $\sin A = \frac{3}{5}$, where $0^\circ < A < 90^\circ$.

$$\therefore \cos^2 A = 1 - \sin^2 A$$

$$\Rightarrow \cos A = +\sqrt{1 - \sin^2 A} = \sqrt{1 - \frac{9}{25}} = \frac{4}{5} \quad [\because \cos A > 0 \text{ for } 0 < A < 90^\circ]$$

$$\therefore \tan A = \frac{\sin A}{\cos A} = \frac{3}{4}$$

Now,

$$\sin 2A = 2 \sin A \cos A = 2 \times \frac{3}{5} \times \frac{4}{5} = \frac{24}{25}$$

$$\cos 2A = 1 - 2 \sin^2 A = 1 - 2 \times \left(\frac{3}{5}\right)^2 = \frac{7}{25}$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A} = \frac{2 \times \frac{3}{4}}{1 - \left(\frac{3}{4}\right)^2} = \frac{\frac{6}{4}}{1 - \frac{9}{16}} = \frac{24}{7} \quad \left[\because \tan A = \frac{3}{4}\right]$$

$$\text{and, } \sin 4A = 2 \sin 2A \cos 2A = 2 \times \frac{24}{25} \times \frac{7}{25} = \frac{336}{625} \quad \left[\because \sin 2A = \frac{24}{25} \text{ and } \cos 2A = \frac{7}{25}\right]$$

EXAMPLE 2 If $\tan \alpha = \frac{1}{7}$, $\sin \beta = \frac{1}{\sqrt{10}}$. Prove that $\alpha + 2\beta = \frac{\pi}{4}$, where $0 < \alpha < \frac{\pi}{2}$ and $0 < \beta < \frac{\pi}{2}$.

SOLUTION In order to prove that $\alpha + 2\beta = \frac{\pi}{4}$, it is sufficient to prove that $\tan(\alpha + 2\beta) = \tan \frac{\pi}{4} = 1$.

We have, $\sin \beta = \frac{1}{\sqrt{10}}$, where $0 < \beta < \frac{\pi}{2}$.

$$\therefore \cos \beta = \sqrt{1 - \sin^2 \beta} = \sqrt{1 - \frac{1}{10}} = \frac{3}{\sqrt{10}} \text{ and, } \tan \beta = \frac{\sin \beta}{\cos \beta} = \frac{\frac{1}{\sqrt{10}}}{\frac{3}{\sqrt{10}}} = \frac{1}{3}$$

In order to find the value of $\tan(\alpha + 2\beta)$, we require the values of $\tan \alpha$ and $\tan 2\beta$. The values of $\tan \alpha$ is given. So, let us find $\tan 2\beta$.

$$\text{Now, } \tan 2\beta = \frac{2 \tan \beta}{1 - \tan^2 \beta}$$

$$\Rightarrow \tan 2\beta = \frac{2 \times \frac{1}{3}}{1 - \frac{1}{9}} = \frac{\frac{2}{3}}{\frac{8}{9}} = \frac{2 \times 9}{3 \times 8} = \frac{3}{4}$$

Thus, we have

$$\tan \alpha = \frac{1}{7} \text{ and } \tan 2\beta = \frac{3}{4}$$

$$\therefore \tan(\alpha + 2\beta) = \frac{\tan \alpha + \tan 2\beta}{1 - \tan \alpha \tan 2\beta} = \frac{\frac{1}{7} + \frac{3}{4}}{1 - \frac{1}{7} \times \frac{3}{4}} = \frac{4 + 21}{28 - 3} = 1$$

$$\Rightarrow \alpha + 2\beta = \frac{\pi}{4}$$

Type II ON PROVING RESULTS AND IDENTITIES BASED UPON THE FOLLOWING FORMULAE:

$$\sin 2\theta = 2 \sin \theta \cos \theta, 1 + \cos 2\theta = 2 \cos^2 \theta, 1 - \cos 2\theta = 2 \sin^2 \theta$$

$$\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}, 1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}, 1 - \cos \theta = 2 \sin^2 \frac{\theta}{2}$$

EXAMPLE 3 Prove that :

$$(i) \frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$$

$$(ii) \frac{\sin 2\theta}{1 - \cos 2\theta} = \cot \theta$$

$$(iii) \frac{1 + \sin 2\theta + \cos 2\theta}{1 + \sin 2\theta - \cos 2\theta} = \cot \theta$$

$$(iv) \frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta} = \tan \frac{\theta}{2}$$

$$(v) \frac{\cos 2\theta}{1 + \sin 2\theta} = \tan \left(\frac{\pi}{4} - \theta \right)$$

$$(vi) \frac{\cos \theta}{1 + \sin \theta} = \tan \left(\frac{\pi}{4} - \frac{\theta}{2} \right)$$

SOLUTION (i) We have, $\sin 2\theta = 2 \sin \theta \cos \theta$ and $1 + \cos 2\theta = 2 \cos^2 \theta$.

$$\therefore \text{LHS} = \frac{\sin 2\theta}{1 + \cos 2\theta} = \frac{2 \sin \theta \cos \theta}{2 \cos^2 \theta} = \tan \theta = \text{RHS}$$

$$(ii) \text{LHS} = \frac{\sin 2\theta}{1 - \cos 2\theta} = \frac{2 \sin \theta \cos \theta}{2 \sin^2 \theta} = \cot \theta = \text{RHS}$$

$$(iii) \text{LHS} = \frac{1 + \sin 2\theta + \cos 2\theta}{1 + \sin 2\theta - \cos 2\theta} = \frac{(1 + \cos 2\theta) + \sin 2\theta}{(1 - \cos 2\theta) + \sin 2\theta}$$

$$= \frac{2 \cos^2 \theta + 2 \sin \theta \cos \theta}{2 \sin^2 \theta + 2 \sin \theta \cos \theta} = \frac{2 \cos \theta (\cos \theta + \sin \theta)}{2 \sin \theta (\cos \theta + \sin \theta)} = \frac{\cos \theta}{\sin \theta} = \cot \theta = \text{RHS}$$

$$(iv) \text{LHS} = \frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta} = \frac{(1 - \cos \theta) + \sin \theta}{(1 + \cos \theta) + \sin \theta}$$

$$= \frac{2 \sin^2 \frac{\theta}{2} + 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2} + 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} = \frac{2 \sin \frac{\theta}{2} \left(\sin \frac{\theta}{2} + \cos \frac{\theta}{2} \right)}{2 \cos \frac{\theta}{2} \left(\sin \frac{\theta}{2} + \cos \frac{\theta}{2} \right)} = \tan \frac{\theta}{2} = \text{RHS}$$

$$(v) \text{LHS} = \frac{\cos 2\theta}{1 + \sin 2\theta}$$

$$\Rightarrow \text{LHS} = \frac{\sin \left(\frac{\pi}{2} - 2\theta \right)}{1 + \cos \left(\frac{\pi}{2} - 2\theta \right)} \quad \left[\because \cos A = \sin \left(\frac{\pi}{2} - A \right), \sin A = \cos \left(\frac{\pi}{2} - A \right) \right]$$

$$\Rightarrow \text{LHS} = \frac{2 \sin \left(\frac{\pi}{4} - \theta \right) \cos \left(\frac{\pi}{4} - \theta \right)}{2 \cos^2 \left(\frac{\pi}{4} - \theta \right)} \quad \left[\because \sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2} \text{ \& } 1 + \cos A = 2 \cos^2 \frac{A}{2} \right]$$

$$\Rightarrow \text{LHS} = \tan\left(\frac{\pi}{4} - \theta\right) = \text{RHS}$$

$$(vi) \quad \text{LHS} = \frac{\cos \theta}{1 + \sin \theta} = \frac{\sin\left(\frac{\pi}{2} - \theta\right)}{1 + \cos\left(\frac{\pi}{2} - \theta\right)} = \frac{2 \sin\left(\frac{\pi}{4} - \frac{\theta}{2}\right) \cos\left(\frac{\pi}{4} - \frac{\theta}{2}\right)}{2 \cos^2\left(\frac{\pi}{4} - \frac{\theta}{2}\right)} = \tan\left(\frac{\pi}{4} - \frac{\theta}{2}\right) = \text{RHS}$$

EXAMPLE 4 Show that: $\sqrt{2 + \sqrt{2 + \sqrt{2 + 2 \cos 8\theta}}} = 2 \cos \theta, 0 < \theta < \frac{\pi}{8}$

SOLUTION We have,

$$\text{LHS} = \sqrt{2 + \sqrt{2 + \sqrt{2(1 + \cos 8\theta)}}}$$

$$\Rightarrow \text{LHS} = \sqrt{2 + \sqrt{2 + \sqrt{2(2 \cos^2 4\theta)}}} \quad \left[\because 1 + \cos 8\theta = 2 \cos^2 \frac{8\theta}{2} = 2 \cos^2 4\theta \right]$$

$$\Rightarrow \text{LHS} = \sqrt{2 + \sqrt{2 + \sqrt{4 \cos^2 4\theta}}}$$

$$\Rightarrow \text{LHS} = \sqrt{2 + \sqrt{2 + 2 \cos 4\theta}}$$

$$\Rightarrow \text{LHS} = \sqrt{2 + \sqrt{2(1 + \cos 4\theta)}}$$

$$\Rightarrow \text{LHS} = \sqrt{2 + \sqrt{2(2 \cos^2 2\theta)}} \quad [\because 1 + \cos 4\theta = 2 \cos^2 2\theta]$$

$$\Rightarrow \text{LHS} = \sqrt{2 + 2 \cos 2\theta} = \sqrt{2(1 + \cos 2\theta)} = \sqrt{2(2 \cos^2 \theta)} = 2 \cos \theta = \text{RHS}$$

EXAMPLE 5 Prove that: $\cos 4x = 1 - 8 \sin^2 x \cos^2 x$.

[NCERT]

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \cos 4x = \cos 2(2x) = 1 - 2 \sin^2 2x = 1 - 2(\sin 2x)^2 \\ &= 1 - 2(2 \sin x \cos x)^2 = 1 - 8 \sin^2 x \cos^2 x = \text{RHS} \end{aligned}$$

EXAMPLE 6 Prove that: $(\cos A + \cos B)^2 + (\sin A - \sin B)^2 = 4 \cos^2\left(\frac{A+B}{2}\right)$.

SOLUTION We have,

$$\text{LHS} = (\cos A + \cos B)^2 + (\sin A - \sin B)^2$$

$$\Rightarrow \text{LHS} = (\cos^2 A + \cos^2 B + 2 \cos A \cos B) + (\sin^2 A + \sin^2 B - 2 \sin A \sin B)$$

$$\Rightarrow \text{LHS} = (\cos^2 A + \sin^2 A) + (\cos^2 B + \sin^2 B) + 2(\cos A \cos B - \sin A \sin B)$$

$$\Rightarrow \text{LHS} = 1 + 1 + 2 \cos(A+B)$$

$$\Rightarrow \text{LHS} = 2 + 2 \cos(A+B)$$

$$\Rightarrow \text{LHS} = 2 \left\{ 1 + \cos(A+B) \right\}$$

$$\Rightarrow \text{LHS} = 2 \times 2 \cos^2\left(\frac{A+B}{2}\right) \quad \left[\because 1 + \cos \theta = 2 \cos^2 \frac{\theta}{2} \right]$$

$$\Rightarrow \text{LHS} = 4 \cos^2\left(\frac{A+B}{2}\right) = \text{RHS.}$$

EXAMPLE 7 Prove that: $\frac{\sec 8\theta - 1}{\sec 4\theta - 1} = \frac{\tan 8\theta}{\tan 2\theta}$

[NCERT EXEMPLAR]

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \frac{\sec 8\theta - 1}{\sec 4\theta - 1} \\ &= \frac{\frac{1}{\cos 8\theta} - 1}{\frac{1}{\cos 4\theta} - 1} = \frac{1 - \cos 8\theta}{\cos 8\theta} \times \frac{\cos 4\theta}{1 - \cos 4\theta} \\ &\Rightarrow \text{LHS} = \frac{2 \sin^2 4\theta}{\cos 8\theta} \times \frac{\cos 4\theta}{2 \sin^2 2\theta} \\ &\Rightarrow \text{LHS} = \frac{(2 \sin 4\theta \cos 4\theta)}{\cos 8\theta} \times \frac{\sin 4\theta}{2 \sin^2 2\theta} \\ &\Rightarrow \text{LHS} = \left(\frac{2 \sin 4\theta \cos 4\theta}{\cos 8\theta} \right) \times \left(\frac{2 \sin 2\theta \cos 2\theta}{2 \sin^2 2\theta} \right) \\ &\Rightarrow \text{LHS} = \left(\frac{\sin 2(4\theta)}{\cos 8\theta} \right) \times \left(\frac{\cos 2\theta}{\sin 2\theta} \right) = \left(\frac{\sin 8\theta}{\cos 8\theta} \right) \times \left(\frac{\cos 2\theta}{\sin 2\theta} \right) = \tan 8\theta \cot 2\theta = \frac{\tan 8\theta}{\tan 2\theta} = \text{RHS} \end{aligned}$$

$$\left[\begin{array}{l} \because 1 - \cos 8\theta = 2 \sin^2 \frac{8\theta}{2} = 2 \sin^2 4\theta \\ \text{and, } 1 - \cos 4\theta = 2 \sin^2 \frac{4\theta}{2} = 2 \sin^2 2\theta \end{array} \right]$$

EXAMPLE 8 Prove that: $\left(1 + \cos \frac{\pi}{8}\right) \left(1 + \cos \frac{3\pi}{8}\right) \left(1 + \cos \frac{5\pi}{8}\right) \left(1 + \cos \frac{7\pi}{8}\right) = \frac{1}{8}$

[NCERT EXEMPLAR]

SOLUTION We observe that

$$\begin{aligned} \cos \frac{7\pi}{8} &= \cos \left(\pi - \frac{\pi}{8} \right) = -\cos \frac{\pi}{8} \quad \text{and,} \quad \cos \frac{5\pi}{8} = \cos \left(\pi - \frac{3\pi}{8} \right) = -\cos \frac{3\pi}{8} \\ \therefore \text{LHS} &= \left(1 + \cos \frac{\pi}{8}\right) \left(1 + \cos \frac{3\pi}{8}\right) \left(1 + \cos \frac{5\pi}{8}\right) \left(1 + \cos \frac{7\pi}{8}\right) \\ &\Rightarrow \text{LHS} = \left(1 + \cos \frac{\pi}{8}\right) \left(1 + \cos \frac{3\pi}{8}\right) \left(1 - \cos \frac{3\pi}{8}\right) \left(1 - \cos \frac{\pi}{8}\right) \\ &\Rightarrow \text{LHS} = \left\{ \left(1 + \cos \frac{\pi}{8}\right) \left(1 - \cos \frac{\pi}{8}\right) \right\} \left\{ \left(1 + \cos \frac{3\pi}{8}\right) \left(1 - \cos \frac{3\pi}{8}\right) \right\} \\ &\Rightarrow \text{LHS} = \left(1 - \cos^2 \frac{\pi}{8}\right) \left(1 - \cos^2 \frac{3\pi}{8}\right) \\ &\Rightarrow \text{LHS} = \sin^2 \frac{\pi}{8} \sin^2 \frac{3\pi}{8} \\ &\Rightarrow \text{LHS} = \frac{1}{4} \left(2 \sin^2 \frac{\pi}{8}\right) \left(2 \sin^2 \frac{3\pi}{8}\right) \\ &\Rightarrow \text{LHS} = \frac{1}{4} \left\{ \left(1 - \cos \frac{\pi}{4}\right) \left(1 - \cos \frac{3\pi}{4}\right) \right\} \\ &\Rightarrow \text{LHS} = \frac{1}{4} \left\{ \left(1 - \frac{1}{\sqrt{2}}\right) \left(1 + \frac{1}{\sqrt{2}}\right) \right\} = \frac{1}{4} \left(1 - \frac{1}{2}\right) = \frac{1}{8} = \text{RHS} \end{aligned}$$

$[\because 2 \sin^2 \theta = 1 - \cos 2\theta]$

EXAMPLE 9 Prove that:

(i) $\cos^4 \frac{\pi}{8} + \cos^4 \frac{3\pi}{8} + \cos^4 \frac{5\pi}{8} + \cos^4 \frac{7\pi}{8} = \frac{3}{2}$

(ii) $\sin^4 \frac{\pi}{8} + \sin^4 \frac{3\pi}{8} + \sin^4 \frac{5\pi}{8} + \sin^4 \frac{7\pi}{8} = \frac{3}{2}$

[NCERT EXEMPLAR]

SOLUTION (i) We know that

$$\frac{7\pi}{8} = \pi - \frac{\pi}{8} \text{ and } \frac{5\pi}{8} = \pi - \frac{3\pi}{8}$$

$$[\because \cos(\pi - \theta) = -\cos \theta]$$

$$\therefore \cos \frac{7\pi}{8} = -\cos \frac{\pi}{8} \text{ and } \cos \frac{5\pi}{8} = -\cos \frac{3\pi}{8}$$

$$\Rightarrow \cos^4 \frac{7\pi}{8} = \cos^4 \frac{\pi}{8} \text{ and } \cos^4 \frac{5\pi}{8} = \cos^4 \frac{3\pi}{8}$$

$$\therefore \text{LHS} = \cos^4 \frac{\pi}{8} + \cos^4 \frac{3\pi}{8} + \cos^4 \frac{5\pi}{8} + \cos^4 \frac{7\pi}{8}$$

$$\Rightarrow \text{LHS} = \cos^4 \frac{\pi}{8} + \cos^4 \frac{3\pi}{8} + \cos^4 \frac{3\pi}{8} + \cos^4 \frac{\pi}{8}$$

$$\Rightarrow \text{LHS} = 2 \cos^4 \frac{\pi}{8} + 2 \cos^4 \frac{3\pi}{8}$$

$$\Rightarrow \text{LHS} = 2 \left\{ \left(\cos^2 \frac{\pi}{8} \right)^2 + \left(\cos^2 \frac{3\pi}{8} \right)^2 \right\}$$

$$\Rightarrow \text{LHS} = 2 \left[\left\{ \frac{1 + \cos \frac{\pi}{4}}{2} \right\}^2 + \left\{ \frac{1 + \cos \frac{3\pi}{4}}{2} \right\}^2 \right]$$

$$\left[\because \frac{1 + \cos 2\theta}{2} = \cos^2 \theta \right]$$

$$\Rightarrow \text{LHS} = \frac{2}{4} \left\{ \left(1 + \cos \frac{\pi}{4} \right)^2 + \left(1 + \cos \frac{3\pi}{4} \right)^2 \right\}$$

$$\Rightarrow \text{LHS} = \frac{2}{4} \left\{ \left(1 + \frac{1}{\sqrt{2}} \right)^2 + \left(1 - \frac{1}{\sqrt{2}} \right)^2 \right\}$$

$$\Rightarrow \text{LHS} = \frac{1}{2} \left\{ \left(1 + \frac{1}{\sqrt{2}} \right)^2 + \left(1 - \frac{1}{\sqrt{2}} \right)^2 \right\} = \frac{1}{2} \left\{ \left(1 + \frac{1}{2} + \sqrt{2} \right) + \left(1 + \frac{1}{2} - \sqrt{2} \right) \right\} = \frac{3}{2} = \text{RHS}$$

(ii) We observe that

$$\sin^4 \frac{7\pi}{8} = \sin^4 \left(\pi - \frac{\pi}{8} \right) = \sin^4 \frac{\pi}{8} \text{ and } \sin^4 \frac{5\pi}{8} = \sin^4 \left(\pi - \frac{3\pi}{8} \right) = \sin^4 \frac{3\pi}{8}$$

$$\therefore \text{LHS} = \sin^4 \frac{\pi}{8} + \sin^4 \frac{3\pi}{8} + \sin^4 \frac{5\pi}{8} + \sin^4 \frac{7\pi}{8}$$

$$\Rightarrow \text{LHS} = \sin^4 \frac{\pi}{8} + \sin^4 \frac{3\pi}{8} + \sin^4 \frac{3\pi}{8} + \sin^4 \frac{\pi}{8}$$

$$\Rightarrow \text{LHS} = 2 \left\{ \sin^4 \frac{\pi}{8} + \sin^4 \frac{3\pi}{8} \right\}$$

$$\Rightarrow \text{LHS} = 2 \left\{ \left(\sin^2 \frac{\pi}{8} \right)^2 + \left(\sin^2 \frac{3\pi}{8} \right)^2 \right\}$$

$$\Rightarrow \text{LHS} = 2 \left[\left\{ \frac{1 - \cos \frac{\pi}{4}}{2} \right\}^2 + \left\{ \frac{1 - \cos \frac{3\pi}{4}}{2} \right\}^2 \right]$$

$$\left[\because \frac{1 - \cos 2\theta}{2} = \sin^2 \theta \right]$$

$$\Rightarrow \text{LHS} = \frac{2}{4} \left\{ \left(1 - \cos \frac{\pi}{4} \right)^2 + \left(1 - \cos \frac{3\pi}{4} \right)^2 \right\}$$

$$\Rightarrow \text{LHS} = \frac{1}{2} \left\{ \left(1 - \frac{1}{\sqrt{2}} \right)^2 + \left(1 + \frac{1}{\sqrt{2}} \right)^2 \right\} = \frac{1}{2} \left\{ \left(1 + \frac{1}{2} - \sqrt{2} \right) + \left(1 + \frac{1}{2} + \sqrt{2} \right) \right\} = \frac{3}{2} = \text{RHS}$$

EXAMPLE 10 Prove that:

$$(i) \cos^2 A + \cos^2 \left(A + \frac{2\pi}{3} \right) + \cos^2 \left(A - \frac{2\pi}{3} \right) = \frac{3}{2}$$

$$(ii) \cos^2 A + \cos^2 \left(A + \frac{\pi}{3} \right) + \cos^2 \left(A - \frac{\pi}{3} \right) = \frac{3}{2}$$

[NCERT EXEMPLAR]

SOLUTION (i) We have,

$$\begin{aligned} \text{LHS} &= \cos^2 A + \cos^2 \left(A + \frac{2\pi}{3} \right) + \cos^2 \left(A - \frac{2\pi}{3} \right) \\ \Rightarrow \text{LHS} &= \frac{1}{2} \left\{ 2 \cos^2 A + 2 \cos^2 \left(A + \frac{2\pi}{3} \right) + 2 \cos^2 \left(A - \frac{2\pi}{3} \right) \right\} \\ \Rightarrow \text{LHS} &= \frac{1}{2} \left[1 + \cos 2A + \left\{ 1 + \cos 2 \left(A + \frac{2\pi}{3} \right) \right\} + \left\{ 1 + \cos 2 \left(A - \frac{2\pi}{3} \right) \right\} \right] \\ \Rightarrow \text{LHS} &= \frac{1}{2} \left[1 + \cos 2A + 1 + \cos \left(2A + \frac{4\pi}{3} \right) + 1 + \cos \left(2A - \frac{4\pi}{3} \right) \right] \\ \Rightarrow \text{LHS} &= \frac{1}{2} \left[3 + \cos 2A + \left\{ \cos \left(2A + \frac{4\pi}{3} \right) + \cos \left(2A - \frac{4\pi}{3} \right) \right\} \right] \\ \Rightarrow \text{LHS} &= \frac{1}{2} \left[3 + \cos 2A + 2 \cos 2A \cos \frac{4\pi}{3} \right] \quad [\because \cos(A+B) + \cos(A-B) = 2 \cos A \cos B] \\ \Rightarrow \text{LHS} &= \frac{1}{2} \left[3 + \cos 2A + 2 (\cos 2A) \left(-\frac{1}{2} \right) \right] = \frac{1}{2} (3 + \cos 2A - \cos 2A) = \frac{3}{2} = \text{RHS} \end{aligned}$$

(ii) We have,

$$\begin{aligned} \text{LHS} &= \cos^2 A + \cos^2 \left(A + \frac{\pi}{3} \right) + \cos^2 \left(A - \frac{\pi}{3} \right) \\ \Rightarrow \text{LHS} &= \frac{1}{2} \left\{ 2 \cos^2 A + 2 \cos^2 \left(A + \frac{\pi}{3} \right) + 2 \cos^2 \left(A - \frac{\pi}{3} \right) \right\} \\ \Rightarrow \text{LHS} &= \frac{1}{2} \left\{ (1 + \cos 2A) + 1 + \cos \left(2A + \frac{2\pi}{3} \right) + 1 + \cos \left(2A - \frac{2\pi}{3} \right) \right\} \\ \Rightarrow \text{LHS} &= \frac{1}{2} \left[3 + \cos 2A + \left\{ \cos \left(2A + \frac{2\pi}{3} \right) + \cos \left(2A - \frac{2\pi}{3} \right) \right\} \right] \\ \Rightarrow \text{LHS} &= \frac{1}{2} \left\{ 3 + \cos 2A + 2 \cos 2A \cos \frac{2\pi}{3} \right\} \quad [\because \cos(A+B) + \cos(A-B) = 2 \cos A \cos B] \\ \Rightarrow \text{LHS} &= \frac{1}{2} \left\{ 3 + \cos 2A + 2 (\cos 2A) \times -\frac{1}{2} \right\} \\ \Rightarrow \text{LHS} &= \frac{1}{2} \{ 3 + \cos 2A - \cos 2A \} = \frac{3}{2} = \text{RHS} \end{aligned}$$

EXAMPLE 11 Prove that:

$$(i) \frac{\sin 5x - 2 \sin 3x + \sin x}{\cos 5x - \cos x} = \tan x$$

[NCERT]

$$(ii) \sin 2x + 2 \sin 4x + \sin 6x = 4 \cos^2 x \sin 4x$$

[NCERT]

$$(iii) \frac{\sin x - \sin 3x}{\sin^2 x - \cos^2 x} = 2 \sin x$$

[NCERT]

SOLUTION (i) We have,

$$\text{LHS} = \frac{\sin 5x - 2 \sin 3x + \sin x}{\cos 5x - \cos x}$$

$$\Rightarrow \text{LHS} = \frac{(\sin 5x + \sin x) - 2 \sin 3x}{\cos 5x - \cos x}$$

$$\Rightarrow \text{LHS} = \frac{2 \sin \left(\frac{5x+x}{2} \right) \cos \left(\frac{5x-x}{2} \right) - 2 \sin 3x}{-2 \sin \left(\frac{5x+x}{2} \right) \sin \left(\frac{5x-x}{2} \right)} = \frac{2 \sin 3x \cos 2x - 2 \sin 3x}{-2 \sin 3x \sin 2x}$$

$$\Rightarrow \text{LHS} = -\frac{2 \sin 3x (1 - \cos 2x)}{-2 \sin 3x \sin 2x} = \frac{1 - \cos 2x}{\sin 2x} = \frac{2 \sin^2 x}{2 \sin x \cos x} = \tan x = \text{RHS}$$

(ii) We have,

$$\text{LHS} = \sin 2x + 2 \sin 4x + \sin 6x$$

$$\Rightarrow \text{LHS} = (\sin 6x + \sin 2x) + 2 \sin 4x$$

$$\Rightarrow \text{LHS} = 2 \sin \left(\frac{6x+2x}{2} \right) \cos \left(\frac{6x-2x}{2} \right) + 2 \sin 4x$$

$$\Rightarrow \text{LHS} = 2 \sin 4x \cos 2x + 2 \sin 4x$$

$$\Rightarrow \text{LHS} = 2 \sin 4x (\cos 2x + 1) = 2 \sin 4x \times 2 \cos^2 x = 4 \cos^2 x \sin 4x = \text{RHS}$$

(iii) We have,

$$\text{LHS} = \frac{\sin x - \sin 3x}{\sin^2 x - \cos^2 x} = \frac{2 \sin \left(\frac{x-3x}{2} \right) \cos \left(\frac{x+3x}{2} \right)}{-(\cos^2 x - \sin^2 x)}$$

$$\Rightarrow \text{LHS} = \frac{2 \sin (-x) \cos 2x}{-\cos 2x} = \frac{-2 \sin x \cos 2x}{-\cos 2x} = 2 \sin x = \text{RHS}$$

EXAMPLE 12 Show that: $2 \sin^2 \beta + 4 \cos (\alpha + \beta) \sin \alpha \sin \beta + \cos 2 (\alpha + \beta) = \cos 2 \alpha$

SOLUTION We have,

[NCERT EXEMPLAR]

$$\text{LHS} = 2 \sin^2 \beta + 4 \cos (\alpha + \beta) \sin \alpha \sin \beta + \cos 2 (\alpha + \beta)$$

$$\Rightarrow \text{LHS} = 2 \sin^2 \beta + 2 \cos (\alpha + \beta) (2 \sin \alpha \sin \beta) + \cos 2 (\alpha + \beta)$$

$$\Rightarrow \text{LHS} = 2 \sin^2 \beta + 2 \cos (\alpha + \beta) [\cos (\alpha - \beta) - \cos (\alpha + \beta)] + \cos 2 (\alpha + \beta)$$

$$\Rightarrow \text{LHS} = 2 \sin^2 \beta + 2 \cos (\alpha + \beta) \cos (\alpha - \beta) - 2 \cos^2 (\alpha + \beta) + \cos 2 (\alpha + \beta)$$

$$\Rightarrow \text{LHS} = 2 \sin^2 \beta + 2 (\cos^2 \alpha - \sin^2 \beta) - 2 \cos^2 (\alpha + \beta) + 2 \cos^2 (\alpha + \beta) - 1$$

$$\Rightarrow \text{LHS} = 2 \sin^2 \beta + 2 \cos^2 \alpha - 2 \sin^2 \beta - 2 \cos^2 (\alpha + \beta) + 2 \cos^2 (\alpha + \beta) - 1$$

$$\Rightarrow \text{LHS} = 2 \cos^2 \alpha - 1 = \cos 2 \alpha = \text{RHS}$$

EXAMPLE 13 Show that $\sqrt{3} \operatorname{cosec} 20^\circ - \sec 20^\circ = 4$.

[NCERT EXEMPLAR]

SOLUTION We have,

$$\text{LHS} = \sqrt{3} \operatorname{cosec} 20^\circ - \sec 20^\circ$$

$$\Rightarrow \text{LHS} = \frac{\sqrt{3}}{\sin 20^\circ} - \frac{1}{\cos 20^\circ}$$

$$\Rightarrow \text{LHS} = \frac{\sqrt{3} \cos 20^\circ - \sin 20^\circ}{\sin 20^\circ \cos 20^\circ}$$

$$\Rightarrow \text{LHS} = \frac{2 \left\{ \frac{\sqrt{3}}{2} \cos 20^\circ - \frac{1}{2} \sin 20^\circ \right\}}{\sin 20^\circ \cos 20^\circ}$$

$$\Rightarrow \text{LHS} = \frac{2 (\sin 60^\circ \cos 20^\circ - \cos 60^\circ \sin 20^\circ)}{\sin 20^\circ \cos 20^\circ}$$

$$\Rightarrow \text{LHS} = \frac{2 \sin (60^\circ - 20^\circ)}{\sin 20^\circ \cos 20^\circ} = \frac{2 \sin 40^\circ}{\sin 20^\circ \cos 20^\circ} = \frac{4 \sin 40^\circ}{2 \sin 20^\circ \cos 20^\circ} = \frac{4 \sin 40^\circ}{\sin 40^\circ} = 4 = \text{RHS}$$

EXAMPLE 14 Prove that: $\tan 4\theta = \frac{4 \tan \theta (1 - \tan^2 \theta)}{1 - 6 \tan^2 \theta + \tan^4 \theta}$ [NCERT]

SOLUTION We have,

$$\text{LHS} = \tan 4\theta = \tan (2(2\theta))$$

$$\Rightarrow \text{LHS} = \frac{2 \tan 2\theta}{1 - \tan^2 2\theta}$$

$$\Rightarrow \text{LHS} = \frac{2 \left(\frac{2 \tan \theta}{1 - \tan^2 \theta} \right)}{1 - \left(\frac{2 \tan \theta}{1 - \tan^2 \theta} \right)^2}$$

$$\Rightarrow \text{LHS} = \frac{4 \tan \theta (1 - \tan^2 \theta)}{(1 - \tan^2 \theta)^2 - 4 \tan^2 \theta} = \frac{4 \tan \theta (1 - \tan^2 \theta)}{1 - 6 \tan^2 \theta + \tan^4 \theta} = \text{RHS}$$

Type III ON FINDING THE VALUES OF $\sin \frac{x}{2}$, $\cos \frac{x}{2}$ AND $\tan \frac{x}{2}$ WHEN VALUES OF $\sin x$ OR $\cos x$ OR $\tan x$ ARE GIVEN

EXAMPLE 15 If $0 \leq x \leq 2\pi$, find $\sin \frac{x}{2}$, $\cos \frac{x}{2}$, and $\tan \frac{x}{2}$, when:

(i) $\tan x = -\frac{4}{3}$, x lies in quadrant II (ii) $\cos x = -\frac{1}{3}$, x lies in quadrant III

(iii) $\sin x = -\frac{1}{2}$, x lies in quadrant IV.

SOLUTION (i) It is given that x lies in IInd quadrant in which $\cos x$ is negative.

$$\therefore \cos x = -\frac{1}{\sqrt{1 + \tan^2 x}} = -\frac{1}{\sqrt{1 + 16/9}} = -\frac{3}{5}$$

It is given that x lies in IInd quadrant.

i.e. $\frac{\pi}{2} < x < \pi$

$$\Rightarrow \frac{\pi}{4} < \frac{x}{2} < \frac{\pi}{2}$$

$$\Rightarrow \frac{x}{2} \text{ lies in first quadrant} \Rightarrow \sin \frac{x}{2}, \cos \frac{x}{2} \text{ and } \tan \frac{x}{2} \text{ are positive}$$

$$\therefore \cos \frac{x}{2} = \pm \sqrt{\frac{1 + \cos x}{2}} \Rightarrow \cos \frac{x}{2} = \sqrt{\frac{1 + \cos x}{2}} = \sqrt{\frac{1 - 3/5}{2}} = \frac{1}{\sqrt{5}}$$

$$\sin \frac{x}{2} = \pm \sqrt{\frac{1 - \cos x}{2}} \Rightarrow \sin \frac{x}{2} = \sqrt{\frac{1 - \cos x}{2}} = \sqrt{\frac{1 + 3/5}{2}} = \frac{2}{\sqrt{5}}$$

and, $\tan \frac{x}{2} = \pm \sqrt{\frac{1 - \cos x}{1 + \cos x}} \Rightarrow \tan \frac{x}{2} = \sqrt{\frac{1 - \cos x}{1 + \cos x}} = \sqrt{\frac{1 + 3/5}{1 - 3/5}} = 2$

(ii) It is given that x lies in the III quadrant.

i.e. $\pi < x < \frac{3\pi}{2}$

$$\Rightarrow \frac{\pi}{2} < \frac{x}{2} < \frac{3\pi}{4}$$

$$\Rightarrow \frac{x}{2} \text{ lies in IInd quadrant} \Rightarrow \cos \frac{x}{2} < 0, \sin \frac{x}{2} > 0 \text{ and } \tan \frac{x}{2} < 0$$

$$\therefore \cos \frac{x}{2} = \pm \sqrt{\frac{1 + \cos x}{2}}$$

$$\Rightarrow \cos \frac{x}{2} = -\sqrt{\frac{1 + \cos x}{2}}$$

$$\left[\because \cos \frac{x}{2} \text{ is -ve} \right]$$

$$\Rightarrow \cos \frac{x}{2} = -\sqrt{\frac{1 - 1/3}{2}} = -\frac{1}{\sqrt{3}}$$

$$\left[\because \cos x = -\frac{1}{3} \right]$$

and, $\sin \frac{x}{2} = \pm \sqrt{\frac{1 - \cos x}{2}}$

$$\Rightarrow \sin \frac{x}{2} = \sqrt{\frac{1 - \cos x}{2}}$$

$$\left[\because \sin \frac{x}{2} > 0 \right]$$

$$\Rightarrow \sin \frac{x}{2} = \sqrt{\frac{1 + 1/3}{2}} = \sqrt{\frac{2}{3}}$$

$$\left[\because \cos x = \frac{1}{3} \right]$$

and, $\tan \frac{x}{2} = \frac{\sin x/2}{\cos x/2} = \sqrt{\frac{2}{3}} \times -\sqrt{3} = -\sqrt{2}$

(iii) It is given that x lies in IVth quadrant in which $\cos x$ is positive.

$$\therefore \sin x = -\frac{1}{2} \Rightarrow \cos x = \sqrt{1 - \sin^2 x} = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$$

Now,

x lies in IVth quadrant

$$\Rightarrow \frac{3\pi}{2} < x < 2\pi$$

$$\Rightarrow \frac{3\pi}{4} < \frac{x}{2} < \pi$$

$$\Rightarrow \frac{x}{2} \text{ lies in IInd quadrant} \Rightarrow \cos \frac{x}{2} < 0, \sin \frac{x}{2} > 0 \text{ and } \tan \frac{x}{2} < 0$$

$$\therefore \cos \frac{x}{2} = \pm \sqrt{\frac{1 + \cos x}{2}}$$

$$\Rightarrow \cos \frac{x}{2} = -\sqrt{\frac{1 + \cos x}{2}}$$

$$\left[\because \cos \frac{x}{2} < 0 \right]$$

$$\Rightarrow \cos \frac{x}{2} = -\sqrt{\frac{1 + \sqrt{3}/2}{2}} = -\frac{\sqrt{2 + \sqrt{3}}}{2}$$

$$\left[\because \cos x = \frac{\sqrt{3}}{2} \right]$$

$$\sin \frac{x}{2} = \pm \sqrt{\frac{1 - \cos x}{2}}$$

$$\Rightarrow \sin \frac{x}{2} = \sqrt{\frac{1 - \cos x}{2}} \quad \left[\because \sin \frac{x}{2} > 0 \right]$$

$$\Rightarrow \sin \frac{x}{2} = \sqrt{\frac{1 - \sqrt{3}/2}{2}} = \frac{\sqrt{2 - \sqrt{3}}}{2} \quad \left[\because \cos x = \frac{\sqrt{3}}{2} \right]$$

$$\text{and, } \tan \frac{x}{2} = \frac{\sin(x/2)}{\cos(x/2)} = \frac{\sqrt{2 - \sqrt{3}}}{2} \times \frac{-2}{\sqrt{2 + \sqrt{3}}} = -\sqrt{\frac{2 - \sqrt{3}}{2 + \sqrt{3}}}$$

EXAMPLE 16 If $\tan x = \frac{3}{4}$, $\pi < x < \frac{3\pi}{2}$, find the values of $\sin \frac{x}{2}$, $\cos \frac{x}{2}$ and $\tan \frac{x}{2}$. [NCERT]

SOLUTION It is given that

$$\tan x = \frac{3}{4} \text{ and, } \pi < x < \frac{3\pi}{2}$$

$$\therefore \cos x = -\frac{1}{\sqrt{1 + \tan^2 x}} \quad \left[\because \pi < x < \frac{3\pi}{2} \therefore \cos x \text{ is negative} \right]$$

$$\Rightarrow \cos x = -\frac{1}{\sqrt{1 + \frac{9}{16}}} = -\frac{4}{5}$$

$$\therefore \pi < x < \frac{3\pi}{2} \Rightarrow \frac{\pi}{2} < \frac{x}{2} < \frac{3\pi}{4} \Rightarrow \cos x < 0 \text{ and } \sin x > 0$$

$$\therefore \cos \frac{x}{2} = -\sqrt{\frac{1 + \cos x}{2}} \text{ and, } \sin \frac{x}{2} = \sqrt{\frac{1 - \cos x}{2}}$$

$$\Rightarrow \cos \frac{x}{2} = -\sqrt{\frac{1 - 4/5}{2}} \text{ and, } \sin \frac{x}{2} = \sqrt{\frac{1 + 4/5}{2}}$$

$$\Rightarrow \cos \frac{x}{2} = -\frac{1}{\sqrt{10}} \text{ and, } \sin \frac{x}{2} = \frac{3}{\sqrt{10}}$$

$$\therefore \tan \frac{x}{2} = \frac{\sin \frac{x}{2}}{\cos \frac{x}{2}} = \frac{3/\sqrt{10}}{-1/\sqrt{10}} = -3$$

$$\text{Hence, } \cos \frac{x}{2} = -\frac{1}{\sqrt{10}}, \sin \frac{x}{2} = \frac{3}{\sqrt{10}} \text{ and } \tan \frac{x}{2} = -3$$

Type IV ON FINDING THE VALUES TRIGONOMETRICAL FUNCTIONS FOR $\frac{\pi}{24}, \frac{\pi}{16}, \frac{\pi}{8}$

$$\text{Formulae: } \cos \frac{A}{2} = \pm \sqrt{\frac{1 + \cos A}{2}}, \sin \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{2}}, \tan \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{1 + \cos A}}$$

EXAMPLE 17 Find the values of

$$(i) \cos \frac{\pi}{8} \quad (ii) \sin \frac{\pi}{8} \quad (iii) \tan \frac{\pi}{8} \quad [\text{NCERT}] \quad (iv) \sin \frac{\pi}{24} \quad (v) \cos \frac{\pi}{24}$$

$$\text{SOLUTION (i) We know that } \cos \frac{A}{2} = \pm \sqrt{\frac{1 + \cos A}{2}}$$

Putting $A = \frac{\pi}{4}$, we get

$$\cos \frac{\pi}{8} = \sqrt{\frac{1 + \cos \pi/4}{2}}$$

$$\left[\because \cos \frac{\pi}{8} \text{ is + ve} \right]$$

$$\Rightarrow \cos \frac{\pi}{8} = \sqrt{\frac{1 + 1/\sqrt{2}}{2}} = \sqrt{\frac{\sqrt{2} + 1}{2\sqrt{2}}}$$

(ii) We have,

$$\sin \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{2}}$$

Putting $A = \frac{\pi}{4}$, we get

$$\sin \frac{\pi}{8} = \sqrt{\frac{1 - \cos \pi/4}{2}}$$

$$\left[\because \sin \frac{\pi}{8} \text{ is + ve} \right]$$

$$\Rightarrow \sin \frac{\pi}{8} = \sqrt{\frac{1 - 1/\sqrt{2}}{2}} = \sqrt{\frac{\sqrt{2} - 1}{2\sqrt{2}}}$$

(iii) We have,

$$\tan \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{1 + \cos A}}$$

Putting $A = \frac{\pi}{4}$, we get

$$\tan \frac{\pi}{8} = \sqrt{\frac{1 - \cos \pi/4}{1 + \cos \pi/4}}$$

$$\left[\because \tan \frac{\pi}{8} \text{ is + ve} \right]$$

$$\Rightarrow \tan \frac{\pi}{8} = \sqrt{\frac{1 - 1/\sqrt{2}}{1 + 1/\sqrt{2}}} = \sqrt{\frac{\sqrt{2} - 1}{\sqrt{2} + 1}} = \sqrt{\frac{(\sqrt{2} - 1)^2}{(\sqrt{2} + 1)(\sqrt{2} - 1)}} = \sqrt{2} - 1$$

(iv) We observe that

$$\cos \frac{\pi}{12} = \cos \left(\frac{\pi}{4} - \frac{\pi}{6} \right) = \cos \frac{\pi}{4} \cos \frac{\pi}{6} + \sin \frac{\pi}{4} \sin \frac{\pi}{6} = \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

Putting $A = \frac{\pi}{12}$ in $\sin \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{2}}$, we get

$$\sin \frac{\pi}{24} = \sqrt{\frac{1 - \cos \frac{\pi}{12}}{2}} = \sqrt{\frac{1 - \frac{\sqrt{3} + 1}{2\sqrt{2}}}{2}} = \sqrt{\frac{2\sqrt{2} - \sqrt{3} - 1}{4\sqrt{2}}} = \sqrt{\frac{4 - \sqrt{6} - \sqrt{2}}{8}} = \sqrt{\frac{4 - \sqrt{6} - \sqrt{2}}{2\sqrt{2}}}$$

(v) Putting $A = \frac{\pi}{12}$ in $\cos \frac{A}{2} = \pm \sqrt{\frac{1 + \cos A}{2}}$, we get

$$\cos \frac{\pi}{24} = \sqrt{\frac{1 + \cos \frac{\pi}{12}}{2}}$$

$$\Rightarrow \cos \frac{\pi}{24} = \sqrt{\frac{1 + \frac{\sqrt{3} + 1}{2\sqrt{2}}}{2}} \quad \left[\because \cos \frac{\pi}{24} = \cos 7\frac{1}{2}^\circ \text{ is + ve} \right]$$

$$\Rightarrow \cos \frac{\pi}{24} = \sqrt{\frac{2\sqrt{2} + \sqrt{3} + 1}{4\sqrt{2}}} = \sqrt{\frac{4 + \sqrt{6} + \sqrt{2}}{8}} = \sqrt{\frac{4 + \sqrt{6} + \sqrt{2}}{2\sqrt{2}}}$$

EXAMPLE 18 Prove that:

$$(i) \cot \frac{\pi}{24} = \sqrt{2} + \sqrt{3} + \sqrt{4} + \sqrt{6}$$

$$(ii) \tan \frac{\pi}{16} = \sqrt{4 + 2\sqrt{2}} - (\sqrt{2} + 1)$$

$$(iii) \tan 142 \frac{1^\circ}{2} = 2 + \sqrt{2} - \sqrt{3} - \sqrt{6}$$

SOLUTION (i) We have,

$$\text{LHS} = \cot \frac{\pi}{24}$$

$$\Rightarrow \text{LHS} = \frac{\cos \frac{\pi}{24}}{\sin \frac{\pi}{24}} = \frac{2 \cos \frac{\pi}{24} \cos \frac{\pi}{24}}{2 \sin \frac{\pi}{24} \cos \frac{\pi}{24}} = \frac{2 \cos^2 \frac{\pi}{24}}{2 \sin \frac{\pi}{24} \cos \frac{\pi}{24}} = \frac{1 + \cos \frac{\pi}{12}}{\sin \frac{\pi}{12}}$$

$$\Rightarrow \text{LHS} = \frac{1 + \cos \left(\frac{\pi}{4} - \frac{\pi}{6} \right)}{\sin \left(\frac{\pi}{4} - \frac{\pi}{6} \right)} = \frac{1 + \frac{\sqrt{3} + 1}{2\sqrt{2}}}{\frac{\sqrt{3} - 1}{2\sqrt{2}}} = \frac{2\sqrt{2} + \sqrt{3} + 1}{\sqrt{3} - 1} = \frac{(2\sqrt{2} + \sqrt{3} + 1)(\sqrt{3} + 1)}{(\sqrt{3} - 1)(\sqrt{3} + 1)}$$

$$\Rightarrow \text{LHS} = \frac{2\sqrt{6} + 2\sqrt{2} + 3 + \sqrt{3} + \sqrt{3} + 1}{3 - 1} = \frac{2\sqrt{6} + 2\sqrt{2} + 2\sqrt{3} + 4}{2}$$

$$\Rightarrow \text{LHS} = \sqrt{2} + \sqrt{3} + 2 + \sqrt{6} = \sqrt{2} + \sqrt{3} + \sqrt{4} + \sqrt{6} = \text{RHS}$$

$$(ii) \text{LHS} = \tan \frac{\pi}{16} = \frac{\sin \frac{\pi}{16}}{\cos \frac{\pi}{16}} = \frac{\sin \frac{\pi}{16}}{\cos \frac{\pi}{16}} \times \frac{2 \sin \frac{\pi}{16}}{2 \sin \frac{\pi}{16}} = \frac{2 \sin^2 \frac{\pi}{16}}{2 \sin \frac{\pi}{16} \cos \frac{\pi}{16}} = \frac{1 - \cos \frac{\pi}{8}}{\sin \frac{\pi}{8}}$$

$$\Rightarrow \text{LHS} = \frac{1 - \sqrt{\frac{1 + \cos \frac{\pi}{4}}{2}}}{\sqrt{\frac{1 - \cos \frac{\pi}{4}}{2}}} \quad \left[\because \cos \frac{\theta}{2} = \sqrt{\frac{1 + \cos \theta}{2}} \text{ and } \sin \frac{\theta}{2} = \sqrt{\frac{1 - \cos \theta}{2}} \right]$$

$$\Rightarrow \text{LHS} = \frac{\sqrt{2} - \sqrt{1 + \cos \frac{\pi}{4}}}{\sqrt{1 - \cos \frac{\pi}{4}}} = \frac{\sqrt{2} - \sqrt{1 + \frac{1}{\sqrt{2}}}}{\sqrt{1 - \frac{1}{\sqrt{2}}}} = \frac{\sqrt{2} - \sqrt{\frac{\sqrt{2} + 1}{\sqrt{2}}}}{\sqrt{\frac{\sqrt{2} - 1}{\sqrt{2}}}}$$

$$\Rightarrow \text{LHS} = \frac{\sqrt{2\sqrt{2}} - \sqrt{\sqrt{2} + 1}}{\sqrt{\sqrt{2} - 1}} = \frac{\sqrt{2\sqrt{2}} - \sqrt{\sqrt{2} + 1}}{\sqrt{\sqrt{2} - 1}} \times \frac{\sqrt{\sqrt{2} + 1}}{\sqrt{\sqrt{2} + 1}}$$

$$\Rightarrow \text{LHS} = \frac{\sqrt{2\sqrt{2}} \sqrt{\sqrt{2} + 1} - \sqrt{(\sqrt{2} + 1)^2}}{\sqrt{(\sqrt{2} + 1)(\sqrt{2} - 1)}} = \frac{\sqrt{2\sqrt{2}}(\sqrt{2} + 1) - (\sqrt{2} + 1)}{2 - 1}$$

$$\Rightarrow \text{LHS} = \sqrt{4 + 2\sqrt{2}} - (\sqrt{2} + 1) = \text{RHS}$$

$$(iii) \text{LHS} = \tan 142 \frac{1^\circ}{2} = \tan \left(180^\circ - 37 \frac{1^\circ}{2} \right) = \tan 37 \frac{1^\circ}{2} = -\tan \frac{5\pi}{24}$$

$$\begin{aligned}
 \Rightarrow \text{LHS} &= -\frac{\sin \frac{5\pi}{24}}{\cos \frac{5\pi}{24}} = -\frac{2\sin^2 \frac{5\pi}{24}}{2\sin \frac{5\pi}{24} \cos \frac{5\pi}{24}} = -\frac{1 - \cos \frac{5\pi}{12}}{\sin \frac{5\pi}{12}} \\
 \Rightarrow \text{LHS} &= -\frac{1 - \cos\left(\frac{\pi}{4} + \frac{\pi}{6}\right)}{\sin\left(\frac{\pi}{4} + \frac{\pi}{6}\right)} = -\frac{1 - \left(\cos \frac{\pi}{4} \cos \frac{\pi}{6} - \sin \frac{\pi}{4} \sin \frac{\pi}{6}\right)}{\sin \frac{\pi}{4} \cos \frac{\pi}{6} + \cos \frac{\pi}{4} \sin \frac{\pi}{6}} \\
 \Rightarrow \text{LHS} &= \left(\frac{1 - \frac{\sqrt{3}-1}{2}}{\frac{\sqrt{3}+1}{2}} \right) = -\left(\frac{2\sqrt{2} - \sqrt{3} + 1}{\sqrt{3} + 1} \right) = -\frac{(2\sqrt{2} - \sqrt{3} + 1)}{(\sqrt{3} + 1)} \times \frac{(\sqrt{3} - 1)}{\sqrt{3} - 1} \\
 \Rightarrow \text{LHS} &= -\left\{ \frac{(2\sqrt{2} - \sqrt{3} + 1)(\sqrt{3} - 1)}{3 - 1} \right\} = -\left\{ \frac{2\sqrt{2}(\sqrt{3} - 1) - (\sqrt{3} - 1)^2}{2} \right\} \\
 \Rightarrow \text{LHS} &= -\left\{ \frac{2\sqrt{2}(\sqrt{3} - 1) - (3 + 1 - 2\sqrt{3})}{2} \right\} = -\left\{ \frac{\sqrt{2}(\sqrt{3} - 1) - (2 - \sqrt{3})}{2} \right\} \\
 \Rightarrow \text{LHS} &= -\sqrt{6} + \sqrt{2} + 2 - \sqrt{3} = 2 + \sqrt{2} - \sqrt{3} - \sqrt{6} = \text{RHS}
 \end{aligned}$$

ILLUSTRATIVE EXAMPLES

LEVEL-2

AN IMPORTANT RESULT Prove that: $\cos A \cos 2A \cos 2^2 A \cos 2^3 A \dots \cos 2^{n-1} A = \frac{\sin 2^n A}{2^n \sin A}$

PROOF $\text{LHS} = \cos A \cos 2A \cos 2^2 A \cos 2^3 A \dots \cos 2^{n-1} A$

$$\begin{aligned}
 &= \frac{1}{2 \sin A} \left\{ (2 \sin A \cos A) \cos 2A \cos 2^2 A \cos 2^3 A \dots \cos 2^{n-1} A \right\} \\
 &= \frac{1}{2 \sin A} \left\{ \sin 2A \cos 2A \cos 2^2 A \cos 2^3 A \dots \cos 2^{n-1} A \right\} \\
 &= \frac{1}{2^2 \sin A} \left\{ (2 \sin 2A \cos 2A) \cos 2^2 A \cos 2^3 A \dots \cos 2^{n-1} A \right\} \\
 &= \frac{1}{2^2 \sin A} \left\{ \sin 2(2A) \cos 2^2 A \cos 2^3 A \dots \cos 2^{n-1} A \right\} \\
 &= \frac{1}{2^3 \sin A} \left\{ (2 \sin 2^2 A \cos 2^2 A) \cos 2^3 A \dots \cos 2^{n-1} A \right\} \\
 &= \frac{1}{2^3 \sin A} \left\{ \sin (2 \times 2^2 A) \cos 2^3 A \dots \cos 2^{n-1} A \right\} \\
 &= \frac{1}{2^3 \sin A} \left\{ \sin 2^3 A \cos 2^3 A \cos 2^4 A \dots \cos 2^{n-1} A \right\} \\
 &= \dots = \frac{1}{2^{n-1} \sin A} \left\{ \sin 2^{n-1} A \cos 2^{n-1} A \right\} = \frac{1}{2^n \sin A} \left\{ 2 \sin 2^{n-1} A \cos 2^{n-1} A \right\} \\
 &= \frac{1}{2^n \sin A} \sin (2 \times 2^{n-1} A) = \frac{1}{2^n \sin A} \sin 2^n A = \text{RHS}
 \end{aligned}$$

Type V PROBLEMS BASED ON THE FORMULA

$$\cos A \cos 2A \cos 2^2 A \cos 2^3 A \dots \cos 2^{n-1} A = \frac{\sin 2^n A}{2^n \sin A}$$

EXAMPLE 19 If $\theta = \frac{\pi}{2^n + 1}$, prove that: $2^n \cos \theta \cos 2\theta \cos 2^2 \theta \dots \cos 2^{n-1} \theta = 1$.

SOLUTION We have,

$$\theta = \frac{\pi}{2^n + 1} \Rightarrow 2^n \theta + \theta = \pi \Rightarrow 2^n \theta = \pi - \theta$$

$$\therefore 2^n \cos \theta \cos 2\theta \cos 2^2 \theta \dots \cos 2^{n-1} \theta$$

$$= 2^n \left\{ \frac{\sin 2^n \theta}{2^n \sin \theta} \right\} = \frac{\sin 2^n \theta}{\sin \theta} = \frac{\sin (\pi - \theta)}{\sin \theta} = 1 \quad [\because 2^n \theta = \pi - \theta]$$

EXAMPLE 20 Prove that: $\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ = \frac{1}{16}$

SOLUTION We have,

$$\text{LHS} = \cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ$$

$$\Rightarrow \text{LHS} = \frac{1}{2} (\cos 20^\circ \cos 40^\circ \cos 80^\circ)$$

$$\Rightarrow \text{LHS} = \frac{1}{2} (\cos A \cos 2A \cos 2^2 A), \text{ where } A = 20^\circ$$

$$\Rightarrow \text{LHS} = \frac{1}{2} \left(\frac{\sin 2^3 A}{2^3 \sin A} \right) = \frac{1}{2^4} \frac{\sin 8A}{\sin A}$$

$$\Rightarrow \text{LHS} = \frac{1}{2^4} \frac{\sin 160^\circ}{\sin 20^\circ} = \frac{1}{2^4} \frac{\sin (180^\circ - 20^\circ)}{\sin 20^\circ} = \frac{1}{2^4} \frac{\sin 20^\circ}{\sin 20^\circ} = \frac{1}{2^4} = \frac{1}{16} = \text{RHS}$$

EXAMPLE 21 Prove that: $\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} = -\frac{1}{8}$.

SOLUTION We have,

$$\text{LHS} = \cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7}$$

$$\Rightarrow \text{LHS} = \cos A \cos 2A \cos 2^2 A, \text{ where } A = \frac{\pi}{7}$$

$$\Rightarrow \text{LHS} = \frac{\sin 2^3 A}{2^3 \sin A}$$

$$\Rightarrow \text{LHS} = \frac{\sin 8 \frac{\pi}{7}}{8 \sin \frac{\pi}{7}} \frac{\sin \left(\pi + \frac{\pi}{7} \right)}{8 \sin \frac{\pi}{7}} = -\frac{\sin \frac{\pi}{7}}{8 \sin \frac{\pi}{7}} = -\frac{1}{8} = \text{RHS}$$

EXAMPLE 22 Prove that: $\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{3\pi}{7} \cos \frac{3\pi}{7} = \frac{1}{8}$

SOLUTION We have,

$$\text{LHS} = \cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{3\pi}{7}$$

$$\Rightarrow \text{LHS} = \cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \left(\pi - \frac{4\pi}{7} \right)$$

$$\Rightarrow \text{LHS} = -\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} = -\left(-\frac{1}{8}\right)$$

[See Example 21]

$$\Rightarrow \text{LHS} = \frac{1}{8} = \text{RHS}$$

EXAMPLE 23 Prove that: $\cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{8\pi}{15} \cos \frac{14\pi}{15} = \frac{1}{16}$

SOLUTION We have,

$$\text{LHS} = \cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{8\pi}{15} \cos \left(\pi - \frac{\pi}{15}\right)$$

$$\Rightarrow \text{LHS} = \left(\cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{8\pi}{15}\right) \left(-\cos \frac{\pi}{15}\right)$$

$$[\because \cos(\pi - \theta) = -\cos \theta]$$

$$\Rightarrow \text{LHS} = -\cos \frac{\pi}{15} \cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{8\pi}{15}$$

$$\Rightarrow \text{LHS} = -\cos A \cos 2A \cos 2^2 A \cos 2^3 A, \text{ where } A = \frac{\pi}{15}$$

$$\Rightarrow \text{LHS} = -\frac{\sin 2^4 A}{2^4 \sin A} = -\frac{\sin 16A}{2^4 \sin A}$$

$$\text{LHS} = -\frac{\sin \frac{16\pi}{15}}{16 \sin \frac{\pi}{15}} = -\frac{\sin \left(\pi + \frac{\pi}{15}\right)}{16 \sin \frac{\pi}{15}} = \frac{\sin \frac{\pi}{15}}{16 \sin \frac{\pi}{15}} = \frac{1}{16} = \text{RHS}$$

EXAMPLE 24 Prove that:

$$\sin \frac{\pi}{14} \sin \frac{3\pi}{14} \sin \frac{5\pi}{14} \sin \frac{7\pi}{14} \sin \frac{9\pi}{14} \sin \frac{11\pi}{14} \sin \frac{13\pi}{14} = \frac{1}{64}$$

SOLUTION We know that

$$A + B = \pi \Rightarrow \sin A = \sin(\pi - B) = \sin B$$

$$\therefore \frac{\pi}{14} + \frac{13\pi}{14} = \pi, \frac{3\pi}{14} + \frac{11\pi}{14} = \pi, \frac{5\pi}{14} + \frac{9\pi}{14} = \pi$$

$$\therefore \sin \frac{\pi}{14} = \sin \frac{13\pi}{14}, \sin \frac{3\pi}{14} = \sin \frac{11\pi}{14}, \sin \frac{5\pi}{14} = \sin \frac{9\pi}{14}$$

$$\therefore \text{LHS} = \sin \frac{\pi}{14} \sin \frac{3\pi}{14} \sin \frac{5\pi}{14} \sin \frac{7\pi}{14} \sin \frac{9\pi}{14} \sin \frac{11\pi}{14} \sin \frac{13\pi}{14}$$

$$\Rightarrow \text{LHS} = \sin \frac{\pi}{14} \sin \frac{3\pi}{14} \sin \frac{5\pi}{14} \sin \frac{7\pi}{14} \sin \frac{3\pi}{14} \sin \frac{\pi}{14}$$

$$\Rightarrow \text{LHS} = \left\{ \sin \frac{\pi}{14} \sin \frac{3\pi}{14} \sin \frac{5\pi}{14} \right\}^2 \times \sin \frac{7\pi}{14}$$

$$\Rightarrow \text{LHS} = \left\{ \sin \frac{\pi}{14} \sin \frac{3\pi}{14} \sin \frac{5\pi}{14} \right\}^2 \times 1$$

$$\Rightarrow \text{LHS} = \left\{ \cos \left(\frac{\pi}{2} - \frac{\pi}{14} \right) \cos \left(\frac{\pi}{2} - \frac{3\pi}{14} \right) \cos \left(\frac{\pi}{2} - \frac{5\pi}{14} \right) \right\}^2$$

$$\Rightarrow \text{LHS} = \left\{ \cos \frac{6\pi}{14} \cos \frac{4\pi}{14} \cos \frac{2\pi}{14} \right\}^2 = \left\{ \cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{3\pi}{7} \right\}^2 = \left(\frac{1}{8} \right)^2 = \frac{1}{64} \quad [\text{See Example 21}]$$

EXAMPLE 25 Find the value of $\sin \frac{\pi}{18} \sin \frac{5\pi}{18} \sin \frac{7\pi}{18}$.

SOLUTION We have,

$$\begin{aligned} & \sin \frac{\pi}{18} \sin \frac{5\pi}{18} \sin \frac{7\pi}{18} \\ &= \cos \left(\frac{\pi}{2} - \frac{\pi}{18} \right) \cos \left(\frac{\pi}{2} - \frac{5\pi}{18} \right) \cos \left(\frac{\pi}{2} - \frac{7\pi}{18} \right) \\ &= \cos \frac{4\pi}{9} \cos \frac{2\pi}{9} \cos \frac{\pi}{9} \\ &= \frac{\sin \left(2^3 \times \frac{\pi}{9} \right)}{2^3 \sin \frac{\pi}{9}} = \frac{\sin \frac{8\pi}{9}}{8 \sin \frac{\pi}{9}} = \frac{\sin \left(\pi - \frac{\pi}{9} \right)}{8 \sin \frac{\pi}{9}} = \frac{\sin \frac{\pi}{9}}{8 \sin \frac{\pi}{9}} = \frac{1}{8} \end{aligned}$$

EXAMPLE 26 Prove that:

$$\cos \frac{\pi}{15} \cos \frac{2\pi}{15} \cos \frac{3\pi}{15} \cos \frac{4\pi}{15} \cos \frac{5\pi}{15} \cos \frac{6\pi}{15} \cos \frac{7\pi}{15} = \frac{1}{128}$$

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \cos \frac{\pi}{15} \cos \frac{2\pi}{15} \cos \frac{3\pi}{15} \cos \frac{4\pi}{15} \cos \frac{5\pi}{15} \cos \frac{6\pi}{15} \cos \frac{7\pi}{15} \\ &= \left\{ \cos \frac{\pi}{15} \cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{7\pi}{15} \right\} \left\{ \cos \frac{3\pi}{15} \cos \frac{6\pi}{15} \right\} \cos \frac{5\pi}{15} \\ &= \frac{1}{2} \left\{ \cos \frac{\pi}{15} \cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{7\pi}{15} \right\} \left\{ \cos \frac{3\pi}{15} \cos \frac{6\pi}{15} \right\} \left[\because \cos \frac{5\pi}{15} = \cos \frac{\pi}{3} = \frac{1}{2} \right] \\ &= -\frac{1}{2} \left\{ \cos \frac{\pi}{15} \cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{8\pi}{15} \right\} \left\{ \cos \frac{3\pi}{15} \cos \frac{6\pi}{15} \right\} \left[\because \cos \frac{7\pi}{15} = \cos \left(\pi - \frac{8\pi}{15} \right) = -\cos \frac{8\pi}{15} \right] \\ &= -\frac{1}{2} \left\{ \frac{\sin \frac{2^4 \pi}{15}}{2^4 \sin \frac{\pi}{15}} \right\} \times \left\{ \frac{\sin \left(2^2 \times \frac{3\pi}{15} \right)}{2^2 \sin \frac{3\pi}{15}} \right\} \\ &= -\frac{1}{2} \left\{ \frac{\sin \frac{16\pi}{15}}{16 \sin \frac{\pi}{15}} \right\} \times \left\{ \frac{\sin \frac{12\pi}{15}}{4 \sin \frac{3\pi}{15}} \right\} = -\frac{1}{2} \left\{ \frac{\sin \left(\pi + \frac{\pi}{15} \right)}{16 \sin \frac{\pi}{15}} \right\} \times \left\{ \frac{\sin \left(\pi - \frac{3\pi}{15} \right)}{4 \sin \frac{3\pi}{15}} \right\} \\ &= -\frac{1}{2} \left\{ \frac{-\sin \frac{\pi}{15}}{16 \sin \frac{\pi}{15}} \right\} \times \left\{ \frac{\sin \frac{3\pi}{15}}{4 \sin \frac{3\pi}{15}} \right\} = -\frac{1}{2} \times -\frac{1}{16} \times \frac{1}{4} = \frac{1}{128} = \text{RHS} \end{aligned}$$

EXAMPLE 27 Prove that: $(1 + \sec 2\theta)(1 + \sec 4\theta)(1 + \sec 8\theta) \dots (1 + \sec 2^n \theta) = \tan 2^n \theta \cot \theta, n \in N$.

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= (1 + \sec 2\theta)(1 + \sec 4\theta)(1 + \sec 8\theta) \dots (1 + \sec 2^n \theta) \\ \Rightarrow \text{LHS} &= \frac{(1 + \cos 2\theta)(1 + \cos 4\theta)(1 + \cos 8\theta) \dots (1 + \cos 2^n \theta)}{\cos 2\theta \cos 4\theta \cos 8\theta \dots \cos 2^n \theta} \\ \Rightarrow \text{LHS} &= \frac{(2 \cos^2 \theta)(2 \cos^2 2\theta)(2 \cos^2 2^2 \theta) \dots (2 \cos^2 2^{n-1} \theta)}{\cos 2\theta \cos 2^2 \theta \cos 2^3 \theta \dots \cos 2^n \theta} \end{aligned}$$

$$\Rightarrow \text{LHS} = \frac{2^n \cos \theta}{\cos 2^n \theta} \left\{ \cos \theta \cos 2\theta \cos 2^2 \theta \dots \cos 2^{n-1} \theta \right\}$$

$$\Rightarrow \text{LHS} = \frac{2^n \cos \theta}{\cos 2^n \theta} \left\{ \frac{\sin 2^n \theta}{2^n \sin \theta} \right\} = \tan 2^n \theta \cot \theta = \text{RHS}$$

Type VI MISCELLANEOUS PROBLEMS BASED UPON FOLLOWING FORMULAE

$$\sin 2\theta = 2 \sin \theta \cos \theta, \cos 2\theta = \cos^2 \theta - \sin^2 \theta, \cos 2\theta = 2 \cos^2 \theta - 1,$$

$$\cos 2\theta = 1 - 2 \sin^2 \theta, \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}, \cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}, \sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$$

EXAMPLE 28 If $\tan^2 \theta = 2 \tan^2 \phi + 1$, prove that $\cos 2\theta + \sin^2 \phi = 0$.

SOLUTION We have,

$$\cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$$

$$\therefore \cos 2\theta + \sin^2 \phi = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} + \sin^2 \phi$$

$$\Rightarrow \cos 2\theta + \sin^2 \phi = \frac{1 - (2 \tan^2 \phi + 1)}{1 + (2 \tan^2 \phi + 1)} + \sin^2 \phi \quad [\because \tan^2 \theta = 2 \tan^2 \phi + 1]$$

$$\Rightarrow \cos 2\theta + \sin^2 \phi = \frac{-2 \tan^2 \phi}{2 + 2 \tan^2 \phi} + \sin^2 \phi = \frac{-\tan^2 \phi}{\sec^2 \phi} + \sin^2 \phi = -\sin^2 \phi + \sin^2 \phi = 0$$

EXAMPLE 29 Prove that:

$$(i) \frac{1 + \cos 4x}{\cot x - \tan x} = \frac{1}{2} \sin 4x$$

$$(ii) \frac{\cos 5x + \cos 4x}{1 - 2 \cos 3x} = -\cos 2x - \cos x$$

$$(iii) \frac{\cos 7x - \cos 8x}{1 + 2 \cos 5x} = \cos 2x - \cos 3x$$

SOLUTION (i) We have,

$$\begin{aligned} \text{LHS} &= \frac{1 + \cos 4x}{\cot x - \tan x} = \frac{2 \cos^2 2x \times \cos x \sin x}{\cos^2 x - \sin^2 x} = \frac{2 \cos^2 2x \times 2 \sin x \cos x}{2 \cos 2x} \\ &= \cos 2x \sin 2x = \frac{1}{2} (2 \sin 2x \cos 2x) = \frac{1}{2} \sin 4x = \text{RHS} \end{aligned}$$

(ii) We have,

$$\text{LHS} = \frac{\cos 5x + \cos 4x}{1 - 2 \cos 3x}$$

$$\Rightarrow \text{LHS} = \frac{\sin 3x (\cos 5x + \cos 4x)}{\sin 3x (1 - 2 \cos 3x)} \quad [\text{Multiplying and dividing by } \sin 3x]$$

$$\Rightarrow \text{LHS} = \frac{\left\{ 2 \sin \frac{3x}{2} \cos \frac{3x}{2} \right\} \left\{ 2 \cos \frac{9x}{2} \cos \frac{x}{2} \right\}}{\sin 3x - 2 \sin 3x \cos 3x}$$

$$\Rightarrow \text{LHS} = \frac{4 \sin \frac{3x}{2} \cos \frac{3x}{2} \cos \frac{9x}{2} \cos \frac{x}{2}}{\sin 3x - \sin 6x} = \frac{4 \sin \frac{3x}{2} \cos \frac{3x}{2} \cos \frac{9x}{2} \cos \frac{x}{2}}{2 \sin \left(\frac{3x - 6x}{2} \right) \cos \left(\frac{3x + 6x}{2} \right)}$$

$$\Rightarrow \text{LHS} = \frac{4 \sin \frac{3x}{2} \cos \frac{3x}{2} \cos \frac{9x}{2} \cos \frac{x}{2}}{2 \sin \left(-\frac{3x}{2}\right) \cos \frac{9x}{2}}$$

$$\Rightarrow \text{LHS} = \frac{4 \sin \frac{3x}{2} \cos \frac{3x}{2} \cos \frac{9x}{2} \cos \frac{x}{2}}{-2 \sin \frac{3x}{2} \cos \frac{9x}{2}} = -2 \cos \frac{3x}{2} \cos \frac{x}{2} = -(\cos 2x + \cos x) = \text{RHS}$$

(iii) We have,

$$\text{LHS} = \frac{\cos 7x - \cos 8x}{1 + 2 \cos 5x}$$

$$\Rightarrow \text{LHS} = \frac{2 \sin \frac{5x}{2} (\cos 7x - \cos 8x)}{2 \sin \frac{5x}{2} (1 + 2 \cos 5x)} \quad \left[\text{Multiplying numerator and denominator by } 2 \sin \frac{5x}{2} \right]$$

$$\Rightarrow \text{LHS} = \frac{2 \sin \frac{5x}{2} \cos 7x - 2 \sin \frac{5x}{2} \cos 8x}{2 \sin \frac{5x}{2} + 4 \sin \frac{5x}{2} \cos 5x} = \frac{2 \sin \frac{5x}{2} \cos 7x - 2 \sin \frac{5x}{2} \cos 8x}{2 \left\{ \sin \frac{5x}{2} + 2 \sin \frac{5x}{2} \cos 5x \right\}}$$

$$\Rightarrow \text{LHS} = \frac{\left(\sin \frac{19x}{2} - \sin \frac{9x}{2} \right) - \left(\sin \frac{21x}{2} - \sin \frac{11x}{2} \right)}{2 \left\{ \sin \frac{5x}{2} + \sin \frac{15x}{2} - \sin \frac{5x}{2} \right\}}$$

$$\Rightarrow \text{LHS} = \frac{\left(\sin \frac{19x}{2} + \sin \frac{11x}{2} \right) - \left(\sin \frac{9x}{2} + \sin \frac{21x}{2} \right)}{2 \sin \frac{15x}{2}}$$

$$\Rightarrow \text{LHS} = \frac{2 \sin \left(\frac{\frac{19x}{2} + \frac{11x}{2}}{2} \right) \cos \left(\frac{\frac{19x}{2} - \frac{11x}{2}}{2} \right) - 2 \sin \left(\frac{\frac{9x}{2} + \frac{21x}{2}}{2} \right) \cos \left(\frac{\frac{9x}{2} - \frac{21x}{2}}{2} \right)}{2 \sin \frac{15x}{2}}$$

$$\Rightarrow \text{LHS} = \frac{2 \sin \frac{15x}{2} \cos 2x - 2 \sin \frac{15x}{2} \cos (-3x)}{2 \sin \frac{15x}{2}} = \cos 2x - \cos 3x = \text{RHS}$$

EXAMPLE 30 If $\tan \alpha = \frac{p}{q}$, where $\alpha = 6\beta$, α being an acute angle, prove that

$$\frac{1}{2} \left\{ p \operatorname{cosec} 2\beta - q \sec 2\beta \right\} = \sqrt{p^2 + q^2}$$

SOLUTION We have, $\tan \alpha = \frac{p}{q}$

$$\therefore \sin \alpha = \frac{p}{\sqrt{p^2 + q^2}} \text{ and, } \cos \alpha = \frac{q}{\sqrt{p^2 + q^2}}$$

Now,

$$\text{LHS} = \frac{1}{2} \left\{ p \operatorname{cosec} 2\beta - q \sec 2\beta \right\}$$

$$\Rightarrow \text{LHS} = \frac{\sqrt{p^2 + q^2}}{2} \left\{ \frac{p}{\sqrt{p^2 + q^2}} \operatorname{cosec} 2\beta - \frac{q}{\sqrt{p^2 + q^2}} \sec 2\beta \right\}$$

$$\Rightarrow \text{LHS} = \frac{\sqrt{p^2 + q^2}}{2} \left\{ \sin \alpha \operatorname{cosec} 2\beta - \cos \alpha \sec 2\beta \right\} = \frac{\sqrt{p^2 + q^2}}{2} \left\{ \frac{\sin \alpha}{\sin 2\beta} - \frac{\cos \alpha}{\cos 2\beta} \right\}$$

$$\Rightarrow \text{LHS} = \frac{\sqrt{p^2 + q^2}}{2} \left\{ \frac{\sin \alpha \cos 2\beta - \cos \alpha \sin 2\beta}{\sin 2\beta \cos 2\beta} \right\}$$

$$\Rightarrow \text{LHS} = \sqrt{p^2 + q^2} \left\{ \frac{\sin (\alpha - 2\beta)}{2 \sin 2\beta \cos 2\beta} \right\} = \sqrt{p^2 + q^2} \left\{ \frac{\sin (6\beta - 2\beta)}{\sin 4\beta} \right\} \quad [\because \alpha = 6\beta]$$

$$\Rightarrow \text{LHS} = \sqrt{p^2 + q^2} = \text{RHS.}$$

EXAMPLE 31 Prove that: $\tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \cot 8\alpha = \cot \alpha$

SOLUTION We have,

$$\cot \theta - \tan \theta = \frac{1}{\tan \theta} - \tan \theta = \frac{1 - \tan^2 \theta}{\tan \theta} = 2 \left\{ \frac{1 - \tan^2 \theta}{2 \tan \theta} \right\} = \frac{2}{\tan 2\theta}$$

$$\Rightarrow \cot \theta - \tan \theta = 2 \cot 2\theta \quad \dots(i)$$

Now,

$$\text{LHS} = \tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \cot 8\alpha$$

$$\Rightarrow \text{LHS} = \cot \alpha - \cot \alpha + \tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \cot 8\alpha$$

$$\Rightarrow \text{LHS} = \cot \alpha - \{ \cot \alpha - \tan \alpha - 2 \tan 2\alpha - 4 \tan 4\alpha - 8 \cot 8\alpha \}$$

$$\Rightarrow \text{LHS} = \cot \alpha - \{ (\cot \alpha - \tan \alpha) - 2 \tan 2\alpha - 4 \tan 4\alpha - 8 \cot 8\alpha \}$$

$$\Rightarrow \text{LHS} = \cot \alpha - \{ 2 \cot 2\alpha - 2 \tan 2\alpha - 4 \tan 4\alpha - 8 \cot 8\alpha \}$$

$$\Rightarrow \text{LHS} = \cot \alpha - \{ 2 (\cot 2\alpha - \tan 2\alpha) - 4 \tan 4\alpha - 8 \cot 8\alpha \} \quad [\text{Using (i)}]$$

$$\Rightarrow \text{LHS} = \cot \alpha - \{ 2 \times 2 \cot 2(2\alpha) - 4 \tan 4\alpha - 8 \cot 8\alpha \} \quad [\text{Using (i)}]$$

$$\Rightarrow \text{LHS} = \cot \alpha - \{ 4 (\cot 4\alpha - \tan 4\alpha) - 8 \cot 8\alpha \} \quad [\text{Using (i)}]$$

$$\Rightarrow \text{LHS} = \cot \alpha - \{ 4 \times 2 \cot 2(4\alpha) - 8 \cot 8\alpha \}$$

$$\Rightarrow \text{LHS} = \cot \alpha - (8 \cot 8\alpha - 8 \cot 8\alpha) = \cot \alpha = \text{RHS}$$

EXAMPLE 32 If $\tan \beta = \frac{\tan \alpha + \tan \gamma}{1 + \tan \alpha \tan \gamma}$, prove that $\sin 2\beta = \frac{\sin 2\alpha + \sin 2\gamma}{1 + \sin 2\alpha \sin 2\gamma}$

SOLUTION It is given that

$$\tan \beta = \frac{\tan \alpha + \tan \gamma}{1 + \tan \alpha \tan \gamma}$$

$$\Rightarrow \tan \beta = \frac{\frac{\sin \alpha}{\cos \alpha} + \frac{\sin \gamma}{\cos \gamma}}{1 + \frac{\sin \alpha}{\cos \alpha} \times \frac{\sin \gamma}{\cos \gamma}} = \frac{\frac{\sin \alpha \cos \gamma + \cos \alpha \sin \gamma}{\cos \alpha \cos \gamma}}{\frac{\cos \alpha \cos \gamma + \sin \alpha \sin \gamma}{\cos \alpha \cos \gamma}} = \frac{\sin (\alpha + \gamma)}{\cos (\alpha - \gamma)} \quad \dots(i)$$

$$\text{Now, } \sin 2\beta = \frac{2 \tan \beta}{1 + \tan^2 \beta}$$

$$\begin{aligned}
 \Rightarrow \sin 2\beta &= \frac{\frac{2 \sin (\alpha + \gamma)}{\cos (\alpha - \gamma)}}{1 + \frac{\sin^2 (\alpha + \gamma)}{\cos^2 (\alpha - \gamma)}} && \text{[Using (i)]} \\
 \Rightarrow \sin 2\beta &= \frac{2 \sin (\alpha + \gamma) \cos (\alpha - \gamma)}{\cos^2 (\alpha - \gamma) + \sin^2 (\alpha + \gamma)} \\
 \Rightarrow \sin 2\beta &= \frac{2 (\sin 2\alpha + \sin 2\gamma)}{2 \cos^2 (\alpha - \gamma) + 2 \sin^2 (\alpha + \gamma)} \\
 \Rightarrow \sin 2\beta &= \frac{2 (\sin 2\alpha + \sin 2\gamma)}{1 + \cos (2\alpha - 2\gamma) + 1 - \cos (2\alpha + 2\gamma)} = \frac{2 (\sin 2\alpha + \sin 2\gamma)}{2 + \cos (2\alpha - 2\gamma) - \cos (2\alpha + 2\gamma)} \\
 \Rightarrow \sin 2\beta &= \frac{2 (\sin 2\alpha + \sin 2\gamma)}{2 + 2 \sin 2\alpha \sin 2\gamma} = \frac{\sin 2\alpha + \sin 2\gamma}{1 + \sin 2\alpha \sin 2\gamma}
 \end{aligned}$$

EXAMPLE 33 If $\sin(\theta + \alpha) = a$ and $\sin(\theta + \beta) = b$, prove that $\cos 2(\alpha - \beta) - 4ab \cos(\alpha - \beta) = 1 - 2a^2 - 2b^2$

SOLUTION We have,

[NCERT EXEMPLAR]

$$\begin{aligned}
 \sin(\theta + \alpha) &= a \text{ and } \sin(\theta + \beta) = b \\
 \therefore \cos(\theta + \alpha) &= \sqrt{1 - \sin^2(\theta + \alpha)} = \sqrt{1 - a^2} \text{ and } \cos(\theta + \beta) = \sqrt{1 - \sin^2(\theta + \beta)} = \sqrt{1 - b^2}
 \end{aligned}$$

Now,

$$\begin{aligned}
 \cos(\alpha - \beta) &= \cos\{(\theta + \alpha) - (\theta + \beta)\} \\
 \Rightarrow \cos(\alpha - \beta) &= \cos(\theta + \alpha) \cos(\theta + \beta) + \sin(\theta + \alpha) \sin(\theta + \beta) \\
 \Rightarrow \cos(\alpha - \beta) &= \sqrt{1 - a^2} \sqrt{1 - b^2} + ab \\
 \Rightarrow \cos(\alpha - \beta) &= ab + \sqrt{1 - a^2 - b^2 + a^2 b^2} \\
 \therefore \cos 2(\alpha - \beta) - 4ab \cos(\alpha - \beta) &= 2 \cos^2(\alpha - \beta) - 1 - 4ab \cos(\alpha - \beta) \\
 &= 2 \left\{ ab + \sqrt{1 - a^2 - b^2 + a^2 b^2} \right\}^2 - 1 - 4ab \left\{ ab + \sqrt{1 - a^2 - b^2 + a^2 b^2} \right\} \\
 &= 2 \left\{ a^2 b^2 + 2ab \sqrt{1 - a^2 - b^2 + a^2 b^2} + 1 - a^2 - b^2 + a^2 b^2 \right\} - 1 - 4a^2 b^2 - 4ab \sqrt{1 - a^2 - b^2 + a^2 b^2} \\
 &= 2 - 2a^2 - 2b^2 - 1 = 1 - 2a^2 - 2b^2.
 \end{aligned}$$

EXAMPLE 34 If $\tan\left(\frac{\pi}{4} + \frac{\theta}{2}\right) = \tan^3\left(\frac{\pi}{4} + \frac{\alpha}{2}\right)$, prove that $\sin \theta = \frac{3 \sin \alpha + \sin^3 \alpha}{1 + 3 \sin^2 \alpha}$.

SOLUTION We have,

$$\begin{aligned}
 \tan\left(\frac{\pi}{4} + \frac{\theta}{2}\right) &= \tan^3\left(\frac{\pi}{4} + \frac{\alpha}{2}\right) \\
 \Rightarrow \frac{1 + \tan \frac{\theta}{2}}{1 - \tan \frac{\theta}{2}} &= \left\{ \frac{1 + \tan \frac{\alpha}{2}}{1 - \tan \frac{\alpha}{2}} \right\}^3 \\
 \Rightarrow \frac{\cos \frac{\theta}{2} + \sin \frac{\theta}{2}}{\cos \frac{\theta}{2} - \sin \frac{\theta}{2}} &= \left\{ \frac{\cos \frac{\alpha}{2} + \sin \frac{\alpha}{2}}{\cos \frac{\alpha}{2} - \sin \frac{\alpha}{2}} \right\}^3
 \end{aligned}$$

$$\Rightarrow \left\{ \frac{\cos \frac{\theta}{2} + \sin \frac{\theta}{2}}{\cos \frac{\theta}{2} - \sin \frac{\theta}{2}} \right\}^2 = \left[\left\{ \frac{\cos \frac{\alpha}{2} + \sin \frac{\alpha}{2}}{\cos \frac{\alpha}{2} - \sin \frac{\alpha}{2}} \right\}^2 \right]^3$$

[On squaring both sides]

$$\Rightarrow \frac{1 + 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{1 - 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} = \left\{ \frac{1 + 2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2}}{1 - 2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2}} \right\}^3$$

$$\Rightarrow \frac{1 + \sin \theta}{1 - \sin \theta} = \left(\frac{1 + \sin \alpha}{1 - \sin \alpha} \right)^3$$

$$\Rightarrow \frac{(1 + \sin \theta) - (1 - \sin \theta)}{(1 + \sin \theta) + (1 - \sin \theta)} = \frac{(1 + \sin \alpha)^3 - (1 - \sin \alpha)^3}{(1 + \sin \alpha)^3 + (1 - \sin \alpha)^3}$$

[Applying componendo-dividendo]

$$\Rightarrow \frac{2 \sin \theta}{2} = \frac{6 \sin \alpha + 2 \sin^3 \alpha}{2 + 6 \sin^2 \alpha}$$

$$\Rightarrow \sin \theta = \frac{3 \sin \alpha + \sin^3 \alpha}{1 + 3 \sin^2 \alpha}$$

EXAMPLE 35 If $\tan \frac{\theta}{2} = \sqrt{\frac{1-e}{1+e}} \tan \frac{\phi}{2}$, prove that $\cos \phi = \frac{\cos \theta - e}{1 - e \cos \theta}$.

SOLUTION We have,

$$\cos \phi = \frac{1 - \tan^2 \frac{\phi}{2}}{1 + \tan^2 \frac{\phi}{2}}$$

$$\Rightarrow \cos \phi = \frac{1 - \frac{1+e}{1-e} \tan^2 \frac{\theta}{2}}{1 + \frac{1+e}{1-e} \tan^2 \frac{\theta}{2}} \quad \left[\because \tan \frac{\theta}{2} = \sqrt{\frac{1-e}{1+e}} \tan \frac{\phi}{2} \Rightarrow \tan \frac{\phi}{2} = \sqrt{\frac{1+e}{1-e}} \tan \frac{\theta}{2} \right]$$

$$\Rightarrow \cos \phi = \frac{(1-e) - (1+e) \tan^2 \frac{\theta}{2}}{(1-e) + (1+e) \tan^2 \frac{\theta}{2}}$$

$$\Rightarrow \cos \phi = \frac{(1 - \tan^2 \theta/2) - e(1 + \tan^2 \theta/2)}{(1 + \tan^2 \theta/2) - e(1 - \tan^2 \theta/2)}$$

$$\Rightarrow \cos \phi = \frac{\frac{1 - \tan^2 \theta/2}{1 + \tan^2 \theta/2} - e}{1 - e \frac{1 - \tan^2 \theta/2}{1 + \tan^2 \theta/2}} \quad \left[\text{Dividing numerator and denominator by } 1 + \tan^2 \frac{\theta}{2} \right]$$

$$\Rightarrow \cos \phi = \frac{\cos \theta - e}{1 - e \cos \theta}$$

$$\left[\because \cos \theta = \frac{1 - \tan^2 \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}} \right]$$

EXAMPLE 36 Prove that:

$$\frac{2 \cos 2^n \theta + 1}{2 \cos \theta + 1} = (2 \cos \theta - 1)(2 \cos 2\theta - 1)(2 \cos 2^2 \theta - 1) \dots (2 \cos 2^{n-1} \theta - 1)$$

SOLUTION RHS

$$\begin{aligned} &= (2 \cos \theta - 1)(2 \cos 2\theta - 1)(2 \cos 2^2 \theta - 1) \dots (2 \cos 2^{n-1} \theta - 1) \\ &= \frac{1}{(2 \cos \theta + 1)} \left\{ (2 \cos \theta + 1)(2 \cos \theta - 1)(2 \cos 2\theta - 1)(2 \cos 2^2 \theta - 1) \dots (2 \cos 2^{n-1} \theta - 1) \right\} \\ &= \frac{1}{(2 \cos \theta + 1)} \left\{ (4 \cos^2 \theta - 1)(2 \cos 2\theta - 1)(2 \cos 2^2 \theta - 1) \dots (2 \cos 2^{n-1} \theta - 1) \right\} \\ &= \frac{1}{(2 \cos \theta + 1)} \left\{ \{2(1 + \cos 2\theta) - 1\} (2 \cos 2\theta - 1)(2 \cos 2^2 \theta - 1) \dots (2 \cos 2^{n-1} \theta - 1) \right\} \\ &= \frac{1}{(2 \cos \theta + 1)} \left\{ (2 \cos 2\theta + 1)(2 \cos 2\theta - 1)(2 \cos 2^2 \theta - 1) \dots (2 \cos 2^{n-1} \theta - 1) \right\} \\ &= \frac{1}{(2 \cos \theta + 1)} \left\{ (4 \cos^2 2\theta - 1)(2 \cos 2^2 \theta - 1)(2 \cos 2^3 \theta - 1) \dots (2 \cos 2^{n-1} \theta - 1) \right\} \\ &= \frac{1}{(2 \cos \theta + 1)} \left\{ \{2(\cos 4\theta + 1) - 1\} (2 \cos 2^2 \theta - 1)(2 \cos 2^3 \theta - 1) \dots (2 \cos 2^{n-1} \theta - 1) \right\} \\ &= \frac{1}{(2 \cos \theta + 1)} \left\{ (2 \cos 2^2 \theta + 1)(2 \cos 2^2 \theta - 1)(2 \cos 2^3 \theta - 1) \dots (2 \cos 2^{n-1} \theta - 1) \right\} \\ &= \frac{1}{(2 \cos \theta + 1)} \left\{ (4 \cos^2 2^2 \theta - 1)(2 \cos 2^3 \theta - 1) \dots (2 \cos 2^{n-1} \theta - 1) \right\} \\ &= \frac{1}{(2 \cos \theta + 1)} \left\{ \{2(1 + \cos 2^3 \theta) - 1\} (2 \cos 2^3 \theta - 1) \dots (2 \cos 2^{n-1} \theta - 1) \right\} \\ &= \frac{1}{(2 \cos \theta + 1)} \left\{ (2 \cos 2^3 \theta + 1)(2 \cos 2^3 \theta - 1) \dots (2 \cos 2^{n-1} \theta - 1) \right\} \\ &\dots \dots \dots \\ &= \frac{1}{(2 \cos \theta + 1)} (2 \cos 2^{n-1} \theta + 1)(2 \cos 2^{n-1} \theta - 1) \\ &= \frac{1}{(2 \cos \theta + 1)} (4 \cos^2 2^{n-1} \theta - 1) \\ &= \frac{1}{(2 \cos \theta + 1)} \{2(\cos 2 \cdot 2^{n-1} \theta + 1) - 1\} \\ &= \frac{1}{(2 \cos \theta + 1)} (2 \cos 2^n \theta + 2 - 1) = \frac{2 \cos 2^n \theta + 1}{2 \cos \theta + 1} = \text{LHS} \end{aligned}$$

Type VII ON CONDITIONAL IDENTITIES

EXAMPLE 37 If $\tan \frac{\theta}{2} = \sqrt{\frac{a-b}{a+b}} \tan \frac{\phi}{2}$, prove that $\cos \theta = \frac{a \cos \phi + b}{a + b \cos \phi}$.

SOLUTION We have,

$$\cos \theta = \frac{1 - \tan^2 \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}}$$

$$\Rightarrow \cos \theta = \frac{1 - \frac{a-b}{a+b} \tan^2 \frac{\phi}{2}}{1 + \frac{a-b}{a+b} \tan^2 \frac{\phi}{2}} \quad \left[\text{Putting } \tan \frac{\theta}{2} = \sqrt{\frac{a-b}{a+b}} \tan \frac{\phi}{2} \right]$$

$$\Rightarrow \cos \theta = \frac{(a+b) - (a-b) \tan^2 \frac{\phi}{2}}{(a+b) + (a-b) \tan^2 \frac{\phi}{2}}$$

$$\Rightarrow \cos \theta = \frac{a \left(1 - \tan^2 \frac{\phi}{2} \right) + b \left(1 + \tan^2 \frac{\phi}{2} \right)}{a \left(1 + \tan^2 \frac{\phi}{2} \right) + b \left(1 - \tan^2 \frac{\phi}{2} \right)}$$

$$\Rightarrow \cos \theta = \frac{a \left(\frac{1 - \tan^2 \frac{\phi}{2}}{1 + \tan^2 \frac{\phi}{2}} \right) + b}{a + b \left(\frac{1 - \tan^2 \frac{\phi}{2}}{1 + \tan^2 \frac{\phi}{2}} \right)} \quad \left[\text{Dividing numerator and denominator by } 1 + \tan^2 \frac{\phi}{2} \right]$$

$$\Rightarrow \cos \theta = \frac{a \cos \phi + b}{a + b \cos \phi}$$

EXAMPLE 38 If $\cos \theta = \cos \alpha \cos \beta$, prove that $\tan \frac{\theta + \alpha}{2} \tan \frac{\theta - \alpha}{2} = \tan^2 \frac{\beta}{2}$.

SOLUTION We have,

$$\cos \theta = \cos \alpha \cos \beta$$

$$\Rightarrow \cos \beta = \frac{\cos \theta}{\cos \alpha}$$

$$\Rightarrow \frac{1 - \tan^2 \frac{\beta}{2}}{1 + \tan^2 \frac{\beta}{2}} = \frac{\cos \theta}{\cos \alpha}$$

$$\Rightarrow \frac{\left(1 - \tan^2 \frac{\beta}{2} \right) + \left(1 + \tan^2 \frac{\beta}{2} \right)}{\left(1 - \tan^2 \frac{\beta}{2} \right) - \left(1 + \tan^2 \frac{\beta}{2} \right)} = \frac{\cos \theta + \cos \alpha}{\cos \theta - \cos \alpha} \quad [\text{Applying componendo - dividendo}]$$

$$\Rightarrow \frac{2}{-2 \tan^2 \frac{\beta}{2}} = \frac{2 \cos \frac{\theta + \alpha}{2} \cos \frac{\theta - \alpha}{2}}{-2 \sin \frac{\theta + \alpha}{2} \sin \frac{\theta - \alpha}{2}}$$

$$\Rightarrow \frac{1}{\tan^2 \frac{\beta}{2}} = \frac{1}{\tan \frac{\theta + \alpha}{2} \tan \frac{\theta - \alpha}{2}} \Rightarrow \tan^2 \frac{\beta}{2} = \tan \frac{\theta + \alpha}{2} \tan \frac{\theta - \alpha}{2}$$

EXAMPLE 39 If $\cos \theta = \frac{\cos \alpha - \cos \beta}{1 - \cos \alpha \cos \beta}$, prove that $\tan \frac{\theta}{2} = \pm \tan \frac{\alpha}{2} \cot \frac{\beta}{2}$.

SOLUTION We have,

$$\begin{aligned}\cos \theta &= \frac{\cos \alpha - \cos \beta}{1 - \cos \alpha \cos \beta} \\ \Rightarrow \frac{1 - \tan^2 \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}} &= \frac{\cos \alpha - \cos \beta}{1 - \cos \alpha \cos \beta} \\ \Rightarrow \frac{\left(1 - \tan^2 \frac{\theta}{2}\right) + \left(1 + \tan^2 \frac{\theta}{2}\right)}{\left(1 - \tan^2 \frac{\theta}{2}\right) - \left(1 + \tan^2 \frac{\theta}{2}\right)} &= \frac{(\cos \alpha - \cos \beta) + (1 - \cos \alpha \cos \beta)}{(\cos \alpha - \cos \beta) - (1 - \cos \alpha \cos \beta)} \\ \Rightarrow \frac{2}{-2 \tan^2 \frac{\theta}{2}} &= \frac{1 + \cos \alpha - \cos \beta - \cos \alpha \cos \beta}{- \{1 - \cos \alpha + \cos \beta - \cos \alpha \cos \beta\}} \\ \Rightarrow \frac{1}{\tan^2 \frac{\theta}{2}} &= \frac{(1 + \cos \alpha)(1 - \cos \beta)}{(1 - \cos \alpha)(1 + \cos \beta)} \\ \Rightarrow \frac{1}{\tan^2 \frac{\theta}{2}} &= \frac{2 \cos^2 \frac{\alpha}{2} \times 2 \sin^2 \frac{\beta}{2}}{2 \sin^2 \frac{\alpha}{2} \times 2 \cos^2 \frac{\beta}{2}} \Rightarrow \tan^2 \frac{\theta}{2} = \tan^2 \frac{\alpha}{2} \cot^2 \frac{\beta}{2} \Rightarrow \tan \frac{\theta}{2} = \pm \tan \frac{\alpha}{2} \cot \frac{\beta}{2}\end{aligned}$$

EXAMPLE 40 If $\cos \theta = \frac{\cos \alpha \cos \beta}{1 - \sin \alpha \sin \beta}$, prove that one value of $\tan \frac{\theta}{2} = \frac{\tan \frac{\alpha}{2} - \tan \frac{\beta}{2}}{1 - \tan \frac{\alpha}{2} \tan \frac{\beta}{2}}$.

SOLUTION We have, $\cos \theta = \frac{\cos \alpha \cos \beta}{1 - \sin \alpha \sin \beta}$

$$\begin{aligned}\text{Now, } \tan^2 \frac{\theta}{2} &= \frac{1 - \cos \theta}{1 + \cos \theta} \\ \Rightarrow \tan^2 \frac{\theta}{2} &= \frac{1 - \frac{\cos \alpha \cos \beta}{1 - \sin \alpha \sin \beta}}{1 + \frac{\cos \alpha \cos \beta}{1 - \sin \alpha \sin \beta}} \\ \Rightarrow \tan^2 \frac{\theta}{2} &= \frac{1 - \sin \alpha \sin \beta - \cos \alpha \cos \beta}{1 - \sin \alpha \sin \beta + \cos \alpha \cos \beta} \\ \Rightarrow \tan^2 \frac{\theta}{2} &= \frac{1 - (\cos \alpha \cos \beta + \sin \alpha \sin \beta)}{1 + (\cos \alpha \cos \beta - \sin \alpha \sin \beta)} \\ \Rightarrow \tan^2 \frac{\theta}{2} &= \frac{1 - \cos(\alpha - \beta)}{1 + \cos(\alpha + \beta)} \\ \Rightarrow \tan^2 \frac{\theta}{2} &= \frac{2 \sin^2 \left(\frac{\alpha - \beta}{2}\right)}{2 \cos^2 \left(\frac{\alpha + \beta}{2}\right)}\end{aligned}$$

$$\Rightarrow \tan \frac{\theta}{2} = \pm \frac{\sin \left(\frac{\alpha - \beta}{2} \right)}{\cos \left(\frac{\alpha + \beta}{2} \right)}$$

$$\Rightarrow \tan \frac{\theta}{2} = \pm \frac{\sin \frac{\alpha}{2} \cos \frac{\beta}{2} - \cos \frac{\alpha}{2} \sin \frac{\beta}{2}}{\cos \frac{\alpha}{2} \cos \frac{\beta}{2} - \sin \frac{\alpha}{2} \sin \frac{\beta}{2}}$$

$$\Rightarrow \tan \frac{\theta}{2} = \pm \frac{\tan \frac{\alpha}{2} - \tan \frac{\beta}{2}}{1 - \tan \frac{\alpha}{2} \tan \frac{\beta}{2}} \quad \left[\text{Dividing numerator and denominator by } \cos \frac{\alpha}{2} \cos \frac{\beta}{2} \right]$$

EXAMPLE 41 If $\sin \alpha + \sin \beta = a$ and $\cos \alpha + \cos \beta = b$, prove that

$$(i) \cos(\alpha - \beta) = \frac{a^2 + b^2 - 2}{2} \quad (ii) \tan \frac{\alpha - \beta}{2} = \pm \sqrt{\frac{4 - a^2 - b^2}{a^2 + b^2}}$$

SOLUTION (i) We have,

$$\sin \alpha + \sin \beta = a \text{ and } \cos \alpha + \cos \beta = b$$

$$\Rightarrow (\sin \alpha + \sin \beta)^2 + (\cos \alpha + \cos \beta)^2 = a^2 + b^2$$

$$\Rightarrow (\cos^2 \alpha + \sin^2 \alpha) + (\cos^2 \beta + \sin^2 \beta) + 2(\cos \alpha \cos \beta + \sin \alpha \sin \beta) = a^2 + b^2$$

$$\Rightarrow 2 + 2 \cos(\alpha - \beta) = a^2 + b^2$$

$$\Rightarrow \cos(\alpha - \beta) = \frac{a^2 + b^2 - 2}{2}$$

$$(ii) \text{ Now, } \tan^2 \frac{\theta}{2} = \frac{1 - \cos \theta}{1 + \cos \theta}$$

$$\Rightarrow \tan^2 \left(\frac{\alpha - \beta}{2} \right) = \frac{1 - \cos(\alpha - \beta)}{1 + \cos(\alpha - \beta)}$$

$$\Rightarrow \tan^2 \left(\frac{\alpha - \beta}{2} \right) = \frac{1 - \frac{a^2 + b^2 - 2}{2}}{1 + \frac{a^2 + b^2 - 2}{2}}$$

$$\Rightarrow \tan^2 \left(\frac{\alpha - \beta}{2} \right) = \frac{4 - a^2 - b^2}{a^2 + b^2}$$

$$\Rightarrow \tan \left(\frac{\alpha - \beta}{2} \right) = \pm \sqrt{\frac{4 - a^2 - b^2}{a^2 + b^2}}$$

EXAMPLE 42 If α and β are distinct roots of $a \cos \theta + b \sin \theta = c$, prove that $\sin(\alpha + \beta) = \frac{2ab}{a^2 + b^2}$.

SOLUTION It is given that α and β are distinct roots of $a \cos \theta + b \sin \theta = c$

$$\therefore a \cos \alpha + b \sin \alpha = c \text{ and } a \cos \beta + b \sin \beta = c$$

$$\Rightarrow (a \cos \alpha + b \sin \alpha) - (a \cos \beta + b \sin \beta) = c - c$$

$$\Rightarrow a(\cos \alpha - \cos \beta) + (b \sin \alpha - \sin \beta) = 0$$

$$\Rightarrow -2a \sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2} + 2b \sin \frac{\alpha - \beta}{2} \cos \frac{\alpha + \beta}{2} = 0$$

$$\Rightarrow 2a \sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2} = 2b \sin \frac{\alpha - \beta}{2} \cos \frac{\alpha + \beta}{2}$$

$$\Rightarrow \tan \frac{\alpha + \beta}{2} = \frac{b}{a} \quad \left[\because \alpha \neq \beta \therefore \sin \frac{\alpha - \beta}{2} \neq 0 \right]$$

$$\therefore \sin(\alpha + \beta) = \frac{2 \tan \frac{\alpha + \beta}{2}}{1 + \tan^2 \frac{\alpha + \beta}{2}} \Rightarrow \sin(\alpha + \beta) = \frac{\frac{2b}{a}}{1 + \frac{b^2}{a^2}} = \frac{2ab}{a^2 + b^2}$$

ALITER We have,

$$a \cos \theta + b \sin \theta = c \quad \dots(i)$$

$$\Rightarrow a \left(\frac{1 - \tan^2 \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}} \right) + b \left(\frac{2 \tan \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}} \right) = c$$

$$\Rightarrow a \left(1 - \tan^2 \frac{\theta}{2} \right) + 2b \tan \frac{\theta}{2} = c \left(1 + \tan^2 \frac{\theta}{2} \right)$$

$$\Rightarrow (c + a) \tan^2 \frac{\theta}{2} - 2b \tan \frac{\theta}{2} + (c - a) = 0 \quad \dots(ii)$$

It is given that α and β are roots of the equation (i). Therefore, $\tan \frac{\alpha}{2}$ and $\tan \frac{\beta}{2}$ are roots of equation (ii).

$$\therefore \tan \frac{\alpha}{2} + \tan \frac{\beta}{2} = \frac{2b}{c + a} \text{ and } \tan \frac{\alpha}{2} \tan \frac{\beta}{2} = \frac{c - a}{c + a} \quad \dots(iii)$$

$$\text{Now, } \tan \left(\frac{\alpha}{2} + \frac{\beta}{2} \right) = \frac{\tan \frac{\alpha}{2} + \tan \frac{\beta}{2}}{1 - \tan \frac{\alpha}{2} \tan \frac{\beta}{2}}$$

$$\Rightarrow \tan \left(\frac{\alpha + \beta}{2} \right) = \frac{2b/c + a}{1 - \frac{c - a}{c + a}} = \frac{b}{c + a} \quad [\text{Using (iii)}]$$

$$\therefore \sin(\alpha + \beta) = \frac{2 \tan \left(\frac{\alpha + \beta}{2} \right)}{1 + \tan^2 \left(\frac{\alpha + \beta}{2} \right)} = \frac{2 \frac{b}{c + a}}{1 + \frac{b^2}{(c + a)^2}} = \frac{2ab}{a^2 + b^2}$$

EXERCISE 9.1

LEVEL-1

Prove that: (1-27)

$$1. \sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}} = \tan \theta$$

$$2. \frac{\sin 2\theta}{1 - \cos 2\theta} = \cot \theta$$

$$3. \frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$$

$$4. \sqrt{2 + \sqrt{2 + 2 \cos 4\theta}} = 2 \cos \theta$$

$$5. \frac{1 - \cos 2\theta + \sin 2\theta}{1 + \cos 2\theta + \sin 2\theta} = \tan \theta$$

$$6. \frac{\sin \theta + \sin 2\theta}{1 + \cos \theta + \cos 2\theta} = \tan \theta$$

$$7. \frac{\cos 2\theta}{1 + \sin 2\theta} = \tan \left(\frac{\pi}{4} - \theta \right)$$

$$8. \frac{\cos \theta}{1 - \sin \theta} = \tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right)$$

$$9. \cos^2 \frac{\pi}{8} + \cos^2 \frac{3\pi}{8} + \cos^2 \frac{5\pi}{8} + \cos^2 \frac{7\pi}{8} = 2$$

$$10. \sin^2 \frac{\pi}{8} + \sin^2 \frac{3\pi}{8} + \sin^2 \frac{5\pi}{8} + \sin^2 \frac{7\pi}{8} = 2$$

$$11. (\cos \alpha + \cos \beta)^2 + (\sin \alpha + \sin \beta)^2 = 4 \cos^2 \left(\frac{\alpha - \beta}{2} \right)$$

$$12. \sin^2 \left(\frac{\pi}{8} + \frac{A}{2} \right) - \sin^2 \left(\frac{\pi}{8} - \frac{A}{2} \right) = \frac{1}{\sqrt{2}} \sin A$$

$$13. 1 + \cos^2 2\theta = 2(\cos^4 \theta + \sin^4 \theta)$$

$$14. \cos^3 2\theta + 3 \cos 2\theta = 4(\cos^6 \theta - \sin^6 \theta)$$

$$15. (\sin 3A + \sin A) \sin A + (\cos 3A - \cos A) \cos A = 0$$

$$16. \cos^2 \left(\frac{\pi}{4} - \theta \right) - \sin^2 \left(\frac{\pi}{4} - \theta \right) = \sin 2\theta$$

$$17. \cos 4A = 1 - 8 \cos^2 A + 8 \cos^4 A$$

$$18. \sin 4A = 4 \sin A \cos^3 A - 4 \cos A \sin^3 A$$

[NCERT EXEMPLAR]

$$19. 3(\sin x - \cos x)^4 + 6(\sin x + \cos x)^2 + 4(\sin^6 x + \cos^6 x) = 13$$

$$20. 2(\sin^6 x + \cos^6 x) - 3(\sin^4 x + \cos^4 x) + 1 = 0$$

$$21. \cos^6 A - \sin^6 A = \cos 2A \left(1 - \frac{1}{4} \sin^2 2A \right)$$

$$22. \tan \left(\frac{\pi}{4} + \theta \right) + \tan \left(\frac{\pi}{4} - \theta \right) = 2 \sec 2\theta$$

$$23. \cot^2 A - \tan^2 A = 4 \cot 2A \operatorname{cosec} 2A$$

$$24. \cos 4\theta - \cos 4\alpha = 8(\cos \theta - \cos \alpha)(\cos \theta + \cos \alpha)(\cos \theta - \sin \alpha)(\cos \theta + \sin \alpha)$$

$$25. \sin 3x + \sin 2x - \sin x = 4 \sin x \cos \frac{x}{2} \cos \frac{3x}{2}$$

[NCERT]

$$26. \tan 82 \frac{1}{2}^\circ = (\sqrt{3} + \sqrt{2})(\sqrt{2} + 1) = \sqrt{2} + \sqrt{3} + \sqrt{4} + \sqrt{6}$$

$$27. \cot 22 \frac{1}{2}^\circ = \sqrt{2} + 1$$

28. (i) If $\cos x = -\frac{3}{5}$ and x lies in the IIIrd quadrant, find the values of $\cos \frac{x}{2}$, $\sin \frac{x}{2}$ and $\sin 2x$.
 (ii) If $\cos x = -\frac{3}{5}$ and x lies in IInd quadrant, find the values of $\sin 2x$ and $\sin \frac{x}{2}$.
29. If $\sin x = \frac{\sqrt{5}}{3}$ and x lies in IInd quadrant, find the values of $\cos \frac{x}{2}$, $\sin \frac{x}{2}$ and $\tan \frac{x}{2}$.
30. (i) If $0 \leq x \leq \pi$ and x lies in the IInd quadrant such that $\sin x = \frac{1}{4}$. Find the values of $\cos \frac{x}{2}$, $\sin \frac{x}{2}$ and $\tan \frac{x}{2}$.
 (ii) If $\cos \theta = \frac{4}{5}$ and θ is acute, find $\tan 2\theta$
 (iii) If $\sin \theta = \frac{4}{5}$ and $0 < \theta < \frac{\pi}{2}$, find the value of $\sin 4\theta$
31. If $\tan x = \frac{b}{a}$, then find the value of $\sqrt{\frac{a+b}{a-b}} + \sqrt{\frac{a-b}{a+b}}$. [NCERT]
32. If $\tan A = \frac{1}{7}$ and $\tan B = \frac{1}{3}$, show that $\cos 2A = \sin 4B$.

LEVEL-2

33. Prove that: $\cos 7^\circ \cos 14^\circ \cos 28^\circ \cos 56^\circ = \frac{\sin 68^\circ}{16 \cos 83^\circ}$
34. Prove that: $\cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{8\pi}{15} \cos \frac{16\pi}{15} = \frac{1}{16}$
35. Prove that: $\cos \frac{\pi}{5} \cos \frac{2\pi}{5} \cos \frac{4\pi}{5} \cos \frac{8\pi}{5} = \frac{-1}{16}$
36. Prove that: $\cos \frac{\pi}{65} \cos \frac{2\pi}{65} \cos \frac{4\pi}{65} \cos \frac{8\pi}{65} \cos \frac{16\pi}{65} \cos \frac{32\pi}{65} = \frac{1}{64}$
37. If $2 \tan \alpha = 3 \tan \beta$, prove that $\tan (\alpha - \beta) = \frac{\sin 2\beta}{5 - \cos 2\beta}$.
38. If $\sin \alpha + \sin \beta = a$ and $\cos \alpha + \cos \beta = b$, prove that
 (i) $\sin (\alpha + \beta) = \frac{2ab}{a^2 + b^2}$ (ii) $\cos (\alpha - \beta) = \frac{a^2 + b^2 - 2}{2}$
39. If $2 \tan \frac{\alpha}{2} = \tan \frac{\beta}{2}$, prove that $\cos \alpha = \frac{3 + 5 \cos \beta}{5 + 3 \cos \beta}$
40. If $\cos \theta = \frac{\cos \alpha + \cos \beta}{1 + \cos \alpha \cos \beta}$, prove that $\tan \frac{\theta}{2} = \pm \tan \frac{\alpha}{2} \tan \frac{\beta}{2}$
41. If $\sec (\theta + \alpha) + \sec (\theta - \alpha) = 2 \sec \theta$, prove that $\cos \theta = \pm \sqrt{2} \cos \frac{\alpha}{2}$
42. If $\cos \alpha + \cos \beta = \frac{1}{3}$ and $\sin \alpha + \sin \beta = \frac{1}{4}$, prove that $\cos \frac{\alpha - \beta}{2} = \pm \frac{5}{24}$.
43. If $\sin \alpha = \frac{4}{5}$ and $\cos \beta = \frac{5}{13}$, prove that $\cos \frac{\alpha - \beta}{2} = \frac{8}{\sqrt{65}}$.

44. If $a \cos 2\theta + b \sin 2\theta = c$ has α and β as its roots, then prove that

$$(i) \tan \alpha + \tan \beta = \frac{2b}{a+c} \quad [\text{NCERT EXEMPLAR}] \quad (ii) \tan \alpha \tan \beta = \frac{c-a}{c+a}$$

$$(iii) \tan(\alpha + \beta) = \frac{b}{a} \quad [\text{NCERT EXEMPLAR}]$$

45. If $\cos \alpha + \cos \beta = 0 = \sin \alpha + \sin \beta$, then prove that $\cos 2\alpha + \cos 2\beta = -2 \cos(\alpha + \beta)$.

[NCERT EXEMPLAR]

ANSWERS

$$28. (i) -\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, \frac{24}{25} \quad (ii) -\frac{24}{25}, \frac{2}{\sqrt{5}} \quad 29. \frac{1}{\sqrt{6}}, \sqrt{\frac{5}{6}}, \sqrt{5}$$

$$30. (i) \sqrt{\frac{4-\sqrt{15}}{8}}, \sqrt{\frac{4+\sqrt{15}}{8}}, 4+\sqrt{15} \quad (ii) \frac{24}{7} \quad (iii) -\frac{336}{625} \quad 31. \frac{2 \cos x}{\sqrt{\cos 2x}}$$

HINTS TO NCERT & SELECTED PROBLEMS

$$\begin{aligned} 25. \text{ LHS} &= \sin 3x + \sin 2x - \sin x \\ &= (\sin 3x - \sin x) + \sin 2x \\ &= 2 \sin x \cos 2x + \sin 2x \\ &= 2 \sin x \cos 2x + 2 \sin x \cos x \\ &= 2 \sin x (\cos 2x + \cos x) = 2 \sin x \left(2 \cos \frac{3x}{2} \cos \frac{x}{2} \right) = 4 \sin x \cos \frac{x}{2} \cos \frac{3x}{2} = \text{RHS} \end{aligned}$$

$$32. \text{ Use: } \cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}, \sin 4B = \frac{2 \tan 2B}{1 + \tan^2 2B}, \text{ where } \tan 2B = \frac{2 \tan B}{1 - \tan^2 B}$$

33. Let $A = 7^\circ$. Then,

$$\begin{aligned} \cos 7^\circ \cos 14^\circ \cos 28^\circ \cos 56^\circ &= \cos A \cos 2A \cos 2^2 A \cos 2^3 A \\ &= \frac{\sin 2^4 A}{2^4 \sin A} = \frac{\sin 16 A}{16 \sin A} = \frac{\sin 112^\circ}{2 \sin 7^\circ} = \frac{\sin 68^\circ}{2 \cos 83^\circ} \end{aligned}$$

34. Let $A = \frac{2\pi}{15}$. Then,

$$\begin{aligned} \cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{8\pi}{15} \cos \frac{16\pi}{15} \\ &= \cos A \cos 2A \cos 2^2 A \cos 2^3 A \\ &= \frac{\sin 2^4 A}{2^4 \sin A} = \frac{\sin 16 A}{16 \sin A} = \frac{\sin \frac{32\pi}{15}}{16 \sin \frac{2\pi}{15}} = \frac{\sin \left(2\pi + \frac{2\pi}{15} \right)}{16 \sin \frac{2\pi}{15}} = \frac{\sin \frac{2\pi}{15}}{16 \sin \frac{2\pi}{15}} = \frac{1}{16} = \text{RHS} \end{aligned}$$

44. We have,

$$a \cos 2\theta + b \sin 2\theta = c \quad \dots (i)$$

$$\Rightarrow a \left(\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \right) + \frac{2b \tan \theta}{1 + \tan^2 \theta} = c$$

$$\Rightarrow (c+a) \tan^2 \theta - 2b \tan \theta + (c-a) = 0 \quad \dots (ii)$$

It is given that α, β are roots of equation (i). Therefore, $\tan \alpha, \tan \beta$ are roots of equation (ii).

$$\therefore \tan \alpha + \tan \beta = \frac{2b}{c+a} \text{ and, } \tan \alpha \tan \beta = \frac{c-a}{c+a}$$

Now,

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \Rightarrow \tan(\alpha + \beta) = \frac{2b/c + a}{1 - \frac{c-a}{c+a}} = \frac{b}{a}.$$

45. We have, $\cos \alpha + \cos \beta = 0$ and $\sin \alpha + \sin \beta = 0$

$$\therefore (\cos \alpha + \cos \beta)^2 - (\sin \alpha + \sin \beta)^2 = 0^2 - 0^2$$

$$\Rightarrow (\cos^2 \alpha - \sin^2 \alpha) + (\cos^2 \beta - \sin^2 \beta) + 2(\cos \alpha \cos \beta - \sin \alpha \sin \beta) = 0$$

$$\Rightarrow \cos 2\alpha + \cos 2\beta + 2 \cos(\alpha + \beta) = 0$$

$$\Rightarrow \cos 2\alpha + \cos 2\beta = -2 \cos(\alpha + \beta)$$

9.5 TRIGONOMETRIC RATIOS OF ANGLE 3A IN TERMS OF ANGLE A

THEOREM For the values of angle A, for which the two sides are meaningful prove that:

$$(i) \sin 3A = 3 \sin A - 4 \sin^3 A \quad (ii) \cos 3A = 4 \cos^3 A - 3 \cos A$$

$$(iii) \tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$$

PROOF (i) We know that

$$\sin(A + B) = \sin A \cos B + \cos A \sin B.$$

$$\Rightarrow \sin(A + 2A) = \sin A \cos 2A + \cos A \sin 2A$$

[Replacing B by 2A]

$$\Rightarrow \sin 3A = \sin A (1 - 2 \sin^2 A) + \cos A (2 \sin A \cos A)$$

$$[\because \cos 2A = 1 - 2 \sin^2 A \text{ \& } \sin 2A = 2 \sin A \cos A]$$

$$\Rightarrow \sin 3A = \sin A - 2 \sin^3 A + 2 \sin A (1 - \sin^2 A)$$

$$\Rightarrow \sin 3A = 3 \sin A - 4 \sin^3 A$$

$$\text{Hence, } \sin 3A = 3 \sin A - 4 \sin^3 A$$

(ii) We know that

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\Rightarrow \cos(A + 2A) = \cos A \cos 2A - \sin A \sin 2A$$

[Replacing B by 2A]

$$\Rightarrow \cos 3A = \cos A \cos 2A - \sin A (2 \sin A \cos A)$$

$$[\because \sin 2A = 2 \sin A \cos A]$$

$$\Rightarrow \cos 3A = \cos A (2 \cos^2 A - 1) - 2 \cos A (1 - \cos^2 A)$$

$$[\because \cos 2A = 2 \cos^2 A - 1]$$

$$\Rightarrow \cos 3A = 4 \cos^3 A - 3 \cos A$$

$$\text{Hence, } \cos 3A = 4 \cos^3 A - 3 \cos A$$

(iii) We know that

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\Rightarrow \tan(A + 2A) = \frac{\tan A + \tan 2A}{1 - \tan A \tan 2A}$$

[Replacing B by 2A]

$$\Rightarrow \tan 3A = \frac{\tan A + \frac{2 \tan A}{1 - \tan^2 A}}{1 - \tan A \times \frac{2 \tan A}{1 - \tan^2 A}}$$

$$\left[\because \tan 2A = \frac{2 \tan A}{1 - \tan^2 A} \right]$$

$$\Rightarrow \tan 3A = \frac{\tan A (1 - \tan^2 A) + 2 \tan A}{1 - \tan^2 A - 2 \tan^2 A} = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$$

$$\text{Hence, } \tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$$

Q.E.D.

REMARK It should be noted the angle on the RHS of these formulae is one third of the angle on LHS.

$$\therefore \sin 60^\circ = 3 \sin 20^\circ - 4 \sin^3 20^\circ, \sin 30^\circ = 3 \sin 10^\circ - 4 \sin^3 10^\circ,$$

$$\cos 120^\circ = 4 \cos^3 40^\circ - 3 \cos 40^\circ \text{ etc.}$$

9.6 TRIGONOMETRIC RATIOS OF ANGLE A IN TERMS OF ANGLE A/3

Replacing A by A/3 in the formulas in the above section, we obtain the following formulae:

$$(i) \sin A = 3 \sin\left(\frac{A}{3}\right) - 4 \sin^3\left(\frac{A}{3}\right)$$

$$(ii) \cos A = 4 \cos^3\left(\frac{A}{3}\right) - 3 \cos\left(\frac{A}{3}\right)$$

$$(iii) \tan A = \frac{3 \tan\left(\frac{A}{3}\right) - \tan^3\left(\frac{A}{3}\right)}{1 - 3 \tan^2\left(\frac{A}{3}\right)}$$

ILLUSTRATIVE EXAMPLES

LEVEL-1

EXAMPLE 1 Prove that : $8 \cos^3 \frac{\pi}{9} - 6 \cos \frac{\pi}{9} = 1$

SOLUTION We have,

$$\text{LHS} = 2 \left(4 \cos^3 \frac{\pi}{9} - 3 \cos \frac{\pi}{9} \right) = 2 \cos \left(3 \times \frac{\pi}{9} \right) = 2 \cos \frac{\pi}{3} = 1 = \text{RHS}$$

EXAMPLE 2 Prove that: $108 \sin \frac{\pi}{18} - 144 \sin^3 \frac{\pi}{18} = 18$

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= 108 \sin \frac{\pi}{18} - 144 \sin^3 \frac{\pi}{18} = 36 \left(3 \sin \frac{\pi}{18} - 4 \sin^3 \frac{\pi}{18} \right) \\ &= 36 \sin \left(3 \times \frac{\pi}{18} \right) = 36 \sin \frac{\pi}{6} = 36 \times \frac{1}{2} = 18 = \text{RHS} \end{aligned}$$

EXAMPLE 3 Prove that: $15 \sin \frac{5\pi}{12} + 15 \cos \frac{5\pi}{12} - 20 \sin^3 \frac{5\pi}{12} - 20 \cos^3 \frac{5\pi}{12} = 0$

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= 15 \sin \frac{5\pi}{12} + 15 \cos \frac{5\pi}{12} - 20 \sin^3 \frac{5\pi}{12} - 20 \cos^3 \frac{5\pi}{12} \\ &= \left(15 \sin \frac{5\pi}{12} - 20 \sin^3 \frac{5\pi}{12} \right) - \left(20 \cos^3 \frac{5\pi}{12} - 15 \cos \frac{5\pi}{12} \right) \\ &= 5 \left(3 \sin \frac{5\pi}{12} - 4 \sin^3 \frac{5\pi}{12} \right) - 5 \left(4 \cos^3 \frac{5\pi}{12} - 3 \cos \frac{5\pi}{12} \right) \\ &= 5 \sin \left(3 \times \frac{5\pi}{12} \right) - 5 \cos \left(3 \times \frac{5\pi}{12} \right) = 5 \sin \frac{5\pi}{4} - 5 \cos \frac{5\pi}{4} = -5 \sin \frac{\pi}{4} + 5 \cos \frac{\pi}{4} = 0 \end{aligned}$$

EXAMPLE 4 Prove that: $\cos 6A = 32 \cos^6 A - 48 \cos^4 A + 18 \cos^2 A - 1$

[NCERT]

SOLUTION We have,

$$\begin{aligned} \Rightarrow \text{LHS} &= \cos 6A \\ \Rightarrow \text{LHS} &= 2 \cos^2 3A - 1 && [\because \cos 2\theta = 2 \cos^2 \theta - 1] \\ \Rightarrow \text{LHS} &= 2 (4 \cos^3 A - 3 \cos A)^2 - 1 \\ \Rightarrow \text{LHS} &= 2 (16 \cos^6 A + 9 \cos^2 A - 24 \cos^4 A) - 1 \\ \Rightarrow \text{LHS} &= 32 \cos^6 A - 48 \cos^4 A + 18 \cos^2 A - 1 = \text{RHS} \end{aligned}$$

EXAMPLE 5 Prove that: $\cos A \cos (60^\circ - A) \cos (60^\circ + A) = \frac{1}{4} \cos 3A$

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \cos A \cos (60^\circ - A) \cos (60^\circ + A) \\ \Rightarrow \text{LHS} &= \cos A (\cos^2 60^\circ - \sin^2 A) \quad [\because \cos (A+B) \cos (A-B) = \cos^2 A - \sin^2 B] \\ \Rightarrow \text{LHS} &= \cos A \left(\frac{1}{4} - \sin^2 A \right) = \cos A \left\{ \frac{1}{4} - (1 - \cos^2 A) \right\} = \cos A \left(-\frac{3}{4} + \cos^2 A \right) \\ \Rightarrow \text{LHS} &= \frac{1}{4} \cos A (-3 + 4 \cos^2 A) = \frac{1}{4} (4 \cos^3 A - 3 \cos A) = \frac{1}{4} \cos 3A = \text{RHS} \end{aligned}$$

EXAMPLE 6 Prove that: $\sin A \sin (60^\circ - A) \sin (60^\circ + A) = \frac{1}{4} \sin 3A$

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \sin A \sin (60^\circ - A) \sin (60^\circ + A) \\ \Rightarrow \text{LHS} &= \sin A (\sin^2 60^\circ - \sin^2 A) \quad [\because \sin (A+B) \sin (A-B) = \sin^2 A - \sin^2 B] \\ \Rightarrow \text{LHS} &= \sin A \left(\frac{3}{4} - \sin^2 A \right) \\ \Rightarrow \text{LHS} &= \frac{1}{4} \sin A (3 - 4 \sin^2 A) \\ \Rightarrow \text{LHS} &= \frac{1}{4} (3 \sin A - 4 \sin^3 A) = \frac{1}{4} \sin 3A. \end{aligned}$$

NOTE Reader is advised to learn the results derived in the above two examples as standard results. The following example is an application of the above results.

EXAMPLE 7 Prove that: $\sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ = \frac{3}{16}$

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \frac{\sqrt{3}}{2} \left\{ \sin 20^\circ \sin (60^\circ - 20^\circ) \sin (60^\circ + 20^\circ) \right\} \\ \Rightarrow \text{LHS} &= \frac{\sqrt{3}}{2} \left\{ (\sin A \sin (60^\circ - A) \sin (60^\circ + A)) \right\}, \text{ where } A = 20^\circ \\ \Rightarrow \text{LHS} &= \frac{\sqrt{3}}{2} \times \frac{1}{4} \sin 3A = \frac{\sqrt{3}}{8} \times \sin 60^\circ = \frac{\sqrt{3}}{8} \times \frac{\sqrt{3}}{2} = \frac{3}{16} = \text{RHS} \end{aligned}$$

EXAMPLE 8 Prove that:

$$(i) \tan A + \tan (60^\circ + A) - \tan (60^\circ - A) = 3 \tan 3A$$

$$(ii) \cot A + \cot (60^\circ + A) - \cot (60^\circ - A) = 3 \cot 3A$$

SOLUTION (i) We have,

$$\begin{aligned} \text{LHS} &= \tan A + \tan (60^\circ + A) - \tan (60^\circ - A) \\ \Rightarrow \text{LHS} &= \tan A + \frac{\sqrt{3} + \tan A}{1 - \sqrt{3} \tan A} - \frac{\sqrt{3} - \tan A}{1 + \sqrt{3} \tan A} \\ \Rightarrow \text{LHS} &= \tan A + \frac{(\sqrt{3} + \tan A)(1 + \sqrt{3} \tan A) - (\sqrt{3} - \tan A)(1 - \sqrt{3} \tan A)}{(1 - \sqrt{3} \tan A)(1 + \sqrt{3} \tan A)} \\ \Rightarrow \text{LHS} &= \tan A + \frac{8 \tan A}{1 - 3 \tan^2 A} \\ \Rightarrow \text{LHS} &= \frac{9 \tan A - 3 \tan^3 A}{1 - 3 \tan^2 A} = 3 \left(\frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A} \right) = 3 \tan 3A = \text{RHS} \end{aligned}$$

(ii) We have,

$$\text{LHS} = \cot A + \cot (60^\circ + A) - \cot (60^\circ - A)$$

$$\Rightarrow \text{LHS} = \frac{1}{\tan A} + \frac{1}{\tan (60^\circ + A)} - \frac{1}{\tan (60^\circ - A)}$$

$$\Rightarrow \text{LHS} = \frac{1}{\tan A} + \frac{1 - \sqrt{3} \tan A}{\sqrt{3} + \tan A} - \frac{1 + \sqrt{3} \tan A}{\sqrt{3} - \tan A}$$

$$\Rightarrow \text{LHS} = \frac{1}{\tan A} + \frac{(1 - \sqrt{3} \tan A)(\sqrt{3} - \tan A) - (1 + \sqrt{3} \tan A)(\sqrt{3} + \tan A)}{(\sqrt{3} + \tan A)(\sqrt{3} - \tan A)}$$

$$\Rightarrow \text{LHS} = \frac{1}{\tan A} - \frac{8 \tan A}{3 - \tan^2 A}$$

$$\Rightarrow \text{LHS} = \frac{3 - 9 \tan^2 A}{3 \tan A - \tan^3 A} = 3 \left(\frac{1 - 3 \tan^2 A}{3 \tan A - \tan^3 A} \right) = \frac{3}{\tan 3A} = 3 \cot 3A = \text{RHS}$$

LEVEL-2

EXAMPLE 9 If $\cos \alpha + \cos \beta + \cos \gamma = 0$, then prove that

$$\cos 3\alpha + \cos 3\beta + \cos 3\gamma = 12 \cos \alpha \cos \beta \cos \gamma$$

SOLUTION $\cos 3\alpha + \cos 3\beta + \cos 3\gamma$

$$= (4 \cos^3 \alpha - 3 \cos \alpha) + (4 \cos^3 \beta - 3 \cos \beta) + (4 \cos^3 \gamma - 3 \cos \gamma)$$

$$= 4 (\cos^3 \alpha + \cos^3 \beta + \cos^3 \gamma) - 3 (\cos \alpha + \cos \beta + \cos \gamma)$$

$$= 4 (\cos^3 \alpha + \cos^3 \beta + \cos^3 \gamma) - 3 \times 0$$

$$= 4 \times 3 \cos \alpha \cos \beta \cos \gamma$$

$$[\because a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = 3abc]$$

$$= 12 \cos \alpha \cos \beta \cos \gamma$$

EXAMPLE 10 Prove that: $\sin 3A \sin^3 A + \cos 3A \cos^3 A = \cos^3 2A$

SOLUTION We know that

$$\sin 3A = 3 \sin A - 4 \sin^3 A \Rightarrow \sin^3 A = \frac{3 \sin A - \sin 3A}{4}$$

Similarly,

$$\cos 3A = 4 \cos^3 A - 3 \cos A \Rightarrow \cos^3 A = \frac{\cos 3A + 3 \cos A}{4}$$

$$\therefore \text{LHS} = \sin 3A \sin^3 A + \cos 3A \cos^3 A$$

$$\Rightarrow \text{LHS} = \sin 3A \left\{ \frac{3 \sin A - \sin 3A}{4} \right\} + \cos 3A \left\{ \frac{\cos 3A + 3 \cos A}{4} \right\}$$

$$\Rightarrow \text{LHS} = \frac{1}{4} \left\{ 3 (\cos A \cos 3A + \sin A \sin 3A) + (\cos^2 3A - \sin^2 3A) \right\}$$

$$\Rightarrow \text{LHS} = \frac{1}{4} \left\{ 3 \cos (3A - A) + \cos 2(3A) \right\}$$

$$\Rightarrow \text{LHS} = \frac{1}{4} \left\{ 3 \cos 2A + \cos 3(2A) \right\}$$

$$\Rightarrow \text{LHS} = \frac{1}{4} \left\{ 3 \cos 2A + (4 \cos^3 2A - 3 \cos 2A) \right\} = \cos^3 2A = \text{RHS}$$

EXAMPLE 11 Prove that: $\cos^3 A + \cos^3 (120^\circ + A) + \cos^3 (240^\circ + A) = \frac{3}{4} \cos 3A$

SOLUTION We know that $\cos 3A = 4 \cos^3 A - 3 \cos A$

$$\therefore \cos^3 A = \frac{1}{4}(\cos 3A + 3 \cos A)$$

Now,

$$\begin{aligned} \text{LHS} &= \frac{1}{4} \left\{ \cos 3A + 3 \cos A \right\} + \frac{1}{4} \left\{ \cos (360^\circ + 3A) + 3 \cos (120^\circ + A) \right\} \\ &\quad + \frac{1}{4} \left\{ \cos (720^\circ + 3A) + 3 \cos (240^\circ + A) \right\} \\ \Rightarrow \text{LHS} &= \frac{1}{4} \left\{ \cos 3A + 3 \cos A \right\} + \frac{1}{4} \left\{ \cos 3A + 3 \cos 120^\circ + A \right\} + \frac{1}{4} \left\{ \cos 3A + 3 \cos (240^\circ + A) \right\} \\ \Rightarrow \text{LHS} &= \frac{3}{4} \cos 3A + \frac{3}{4} \left\{ \cos A + \cos (120^\circ + A) + \cos (240^\circ + A) \right\} \\ \Rightarrow \text{LHS} &= \frac{3}{4} \cos 3A + \frac{3}{4} \left\{ \cos A + 2 \cos (180^\circ + A) \cos 60^\circ \right\} \\ \Rightarrow \text{LHS} &= \frac{3}{4} \cos 3A + \frac{3}{4} \left\{ \cos A - 2 \cos A \times \frac{1}{2} \right\} = \frac{3}{4} \cos 3A = \text{RHS} \end{aligned}$$

ALITER We have,

$$\begin{aligned} &\cos A + \cos (120^\circ + A) + \cos (240^\circ + A) \\ &= \cos A + 2 \cos \left(\frac{240^\circ + A + 120^\circ + A}{2} \right) \cos \left(\frac{240^\circ + A - 120^\circ - A}{2} \right) \\ &= \cos A + 2 \cos (180^\circ + A) \cos 60^\circ = \cos A - 2 (\cos A) \times \frac{1}{2} \\ &= \cos A - \cos A = 0 \\ \therefore \cos^3 A + \cos^3 (120^\circ + A) + \cos^3 (240^\circ + A) \\ &= 3 \cos A \cos (120^\circ + A) \cos (240^\circ + A) \quad [\because a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 = 3abc] \\ &= 3 \cos A \cos (180^\circ - 60^\circ + A) \cos (180^\circ + 60^\circ + A) \\ &= 3 \cos A \cos [180^\circ - (60^\circ - A)] \cos [180^\circ + (60^\circ + A)] \\ &= (3 \cos A) \{-\cos (60^\circ - A)\} \{-\cos (60^\circ + A)\} \\ &= 3 \cos A \cos (60^\circ - A) \cos (60^\circ + A) = 3 \times \frac{1}{4} \cos 3A = \frac{3}{4} \cos 3A \end{aligned}$$

EXAMPLE 12 Prove that $\frac{\tan 3x}{\tan x}$ never lies between $\frac{1}{3}$ and 3.

SOLUTION Let $y = \frac{\tan 3x}{\tan x}$. Then,

$$\begin{aligned} y &= \frac{3 \tan x - \tan^3 x}{\tan x (1 - 3 \tan^2 x)} \\ \Rightarrow y &= \frac{3 - \tan^2 x}{1 - 3 \tan^2 x} \Rightarrow (3y - 1) \tan^2 x = y - 3 \Rightarrow \tan^2 x = \frac{y - 3}{3y - 1} \end{aligned}$$

But, $\tan^2 x \geq 0$ for all x



Fig. 9.1 Signs of $\frac{y-3}{3y-1}$ for different values of y

$$\therefore \frac{y-3}{3y-1} \geq 0$$

$$\Rightarrow y < \frac{1}{3} \text{ or, } y \geq 3$$

$\Rightarrow y$ does not lie between $1/3$ and 3 .

Hence, $\frac{\tan 3x}{\tan x}$ never lies between $\frac{1}{3}$ and 3 .

EXAMPLE 13 Prove that: $\cos 5A = 16 \cos^5 A - 20 \cos^3 A + 5 \cos A$

SOLUTION We have,

$$\begin{aligned} \cos 5A &= \cos (3A + 2A) = \cos 3A \cos 2A - \sin 3A \sin 2A \\ \Rightarrow \cos 5A &= (4 \cos^3 A - 3 \cos A)(2 \cos^2 A - 1) - (3 \sin A - 4 \sin^3 A)(2 \sin A \cos A) \\ \Rightarrow \cos 5A &= (4 \cos^3 A - 3 \cos A)(2 \cos^2 A - 1) - (3 - 4 \sin^2 A)(2 \sin^2 A \cos A) \\ \Rightarrow \cos 5A &= (4 \cos^3 A - 3 \cos A)(2 \cos^2 A - 1) - \{3 - 4(1 - \cos^2 A)\} 2(1 - \cos^2 A) \cos A \\ \Rightarrow \cos 5A &= (8 \cos^5 A - 10 \cos^3 A + 3 \cos A) - 2 \cos A (1 - \cos^2 A)(4 \cos^2 A - 1) \\ \Rightarrow \cos 5A &= (8 \cos^5 A - 10 \cos^3 A + 3 \cos A) - 2 \cos A (5 \cos^2 A - 4 \cos^4 A - 1) \\ \Rightarrow \cos 5A &= 16 \cos^5 A - 20 \cos^3 A + 5 \cos A = \text{RHS} \end{aligned}$$

EXERCISE 9.2

LEVEL-1

Prove the following identities (1–8)

- $\sin 5\theta = 5 \sin \theta - 20 \sin^3 \theta + 16 \sin^5 \theta$
- $4(\cos^3 10^\circ + \sin^3 20^\circ) = 3(\cos 10^\circ + \sin 20^\circ)$
- $\cos^3 \theta \sin 3\theta + \sin^3 \theta \cos 3\theta = \frac{3}{4} \sin 4\theta$
- $\sin 5A = 5 \cos^4 A \sin A - 10 \cos^2 A \sin^3 A + \sin^5 A$
- $\tan \theta \tan (\theta + 60^\circ) + \tan \theta \tan (\theta - 60^\circ) + \tan (\theta + 60^\circ) \tan (\theta - 60^\circ) = -3$
- $\tan A + \tan (60^\circ + A) - \tan (60^\circ - A) = 3 \tan 3A$
- $\cot A + \cot (60^\circ + A) - \cot (60^\circ - A) = 3 \cot 3A$
- $\cot A + \cot (60^\circ + A) + \cot (120^\circ + A) = 3 \cot 3A$

LEVEL-2

- Prove that: $\sin^3 A + \sin^3 \left(\frac{2\pi}{3} + A \right) + \sin^3 \left(\frac{4\pi}{3} + A \right) = -\frac{3}{4} \sin 3A$.
- Prove that: $|\sin \theta \sin (60^\circ - \theta) \sin (60^\circ + \theta)| \leq \frac{1}{4}$ for all values of θ .
- Prove that: $|\cos \theta \cos (60^\circ - \theta) \cos (60^\circ + \theta)| \leq \frac{1}{4}$ for all values of θ .

HINTS TO SELECTED PROBLEMS

- We have, $|\sin \theta \sin (60^\circ - \theta) \sin (60^\circ + \theta)| = \frac{1}{4} |\sin 3\theta| \leq \frac{1}{4}$ [$\because |\sin 3\theta| \leq 1$]
- We have, $|\cos \theta \cos (60^\circ - \theta) \cos (60^\circ + \theta)| = \frac{1}{4} |\cos 3\theta| \leq \frac{1}{4}$

9.7 TRIGONOMETRICAL RATIOS OF SOME IMPORTANT ANGLES

By using the formulae introduced in the previous sections we can now find the trigonometrical ratios of some important angles of degree measures 18° , 36° , 54° etc.

THEOREM 1 Prove that: $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$.

PROOF Let $\theta = 18^\circ$. Then,

$$5\theta = 90^\circ$$

$$\Rightarrow 2\theta + 3\theta = 90^\circ$$

$$\Rightarrow 2\theta = 90^\circ - 3\theta$$

$$\Rightarrow \sin 2\theta = \sin (90^\circ - 3\theta)$$

$$\Rightarrow \sin 2\theta = \cos 3\theta$$

$$\Rightarrow 2 \sin \theta \cos \theta = 4 \cos^3 \theta - 3 \cos \theta$$

$$\Rightarrow \cos \theta (2 \sin \theta - 4 \cos^2 \theta + 3) = 0$$

$$\Rightarrow 2 \sin \theta - 4 \cos^2 \theta + 3 = 0$$

$$[\because \cos \theta = \cos 18^\circ \neq 0]$$

$$\Rightarrow 2 \sin \theta - 4(1 - \sin^2 \theta) + 3 = 0$$

$$\Rightarrow 4 \sin^2 \theta + 2 \sin \theta - 1 = 0$$

$$\Rightarrow \sin \theta = \frac{-2 \pm \sqrt{4+16}}{8}$$

$$\Rightarrow \sin \theta = \frac{-1 \pm \sqrt{5}}{4}$$

$$\Rightarrow \sin \theta = \frac{-1 + \sqrt{5}}{4} = \frac{\sqrt{5}-1}{4}$$

$$[\because \theta \text{ lies in Ist quadrant } \therefore \sin \theta > 0]$$

$$\text{Hence, } \sin 18^\circ = \frac{\sqrt{5}-1}{4}$$

Q.E.D.

THEOREM 2 Prove that: $\cos 18^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4}$.

PROOF Putting $\theta = 18^\circ$ in $\cos \theta = \sqrt{1 - \sin^2 \theta}$, we get

$$\cos 18^\circ = \sqrt{1 - \sin^2 18^\circ} = \sqrt{1 - \left(\frac{\sqrt{5}-1}{4}\right)^2} = \sqrt{\frac{16 - (5 + 1 - 2\sqrt{5})}{16}} = \frac{\sqrt{10+2\sqrt{5}}}{4}$$

$$\text{Hence, } \cos 18^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4}$$

Q.E.D.

REMARK The complement of 18° is 72° .

$$\therefore \sin 72^\circ = \sin (90^\circ - 18^\circ) = \cos 18^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4}$$

$$\text{and, } \cos 72^\circ = \cos (90^\circ - 18^\circ) = \sin 18^\circ = \frac{\sqrt{5}-1}{4}$$

The remaining trigonometrical ratios of 18° may be obtained from the above values.

THEOREM 3 Prove that: $\cos 36^\circ = \frac{\sqrt{5}+1}{4}$.

PROOF We have, $\cos 2\theta = 1 - 2 \sin^2 \theta$

$$\therefore \cos 36^\circ = 1 - 2 \sin^2 18^\circ$$

[Putting $\theta = 18^\circ$]

$$\Rightarrow \cos 36^\circ = 1 - 2 \left(\frac{\sqrt{5} - 1}{4} \right)^2 = 1 - 2 \left(\frac{6 - 2\sqrt{5}}{16} \right) = 1 - \left(\frac{3 - \sqrt{5}}{4} \right) = \frac{\sqrt{5} + 1}{4}$$

$$\text{Hence, } \cos 36^\circ = \frac{\sqrt{5} + 1}{4}$$

Q.E.D.

THEOREM 4 Prove that: $\sin 36^\circ = \frac{\sqrt{10 - 2\sqrt{5}}}{4}$.

PROOF Putting $\theta = 36^\circ$ in $\sin \theta = \sqrt{1 - \cos^2 \theta}$, we obtain

$$\sin 36^\circ = \sqrt{1 - \cos^2 36^\circ} = \sqrt{1 - \left(\frac{\sqrt{5} + 1}{4} \right)^2} = \sqrt{\frac{16 - (6 + 2\sqrt{5})}{16}} = \frac{\sqrt{10 - 2\sqrt{5}}}{4}$$

$$\text{Hence, } \sin 36^\circ = \frac{\sqrt{10 - 2\sqrt{5}}}{4}$$

Q.E.D.

REMARK The complement of 36° is 54° .

$$\therefore \sin 54^\circ = \sin (90^\circ - 36^\circ) = \cos 36^\circ = \frac{\sqrt{5} + 1}{4} \text{ and, } \cos 54^\circ = \cos (90^\circ - 36^\circ) = \sin 36^\circ = \frac{\sqrt{10 - 2\sqrt{5}}}{4}$$

The other trigonometrical ratios of 36° may be obtained from the above values.

ILLUSTRATIVE EXAMPLES

LEVEL-1

EXAMPLE 1 Prove that:

$$(i) \sin^2 72^\circ - \sin^2 60^\circ = \frac{\sqrt{5} - 1}{8}$$

$$(ii) \cos^2 48^\circ - \sin^2 12^\circ = \frac{\sqrt{5} + 1}{8}$$

$$(iii) \sin \frac{\pi}{10} + \sin \frac{13\pi}{10} = -\frac{1}{2}$$

$$(iv) \sin \frac{\pi}{10} \sin \frac{13\pi}{10} = -\frac{1}{4}$$

$$(v) \sin^2 24^\circ - \sin^2 6^\circ = \frac{\sqrt{5} - 1}{8}$$

SOLUTION (i) We have,

$$\text{LHS} = \sin^2 72^\circ - \sin^2 60^\circ$$

$$\Rightarrow \text{LHS} = \cos^2 18^\circ - \sin^2 60^\circ$$

$$[\because \sin 72^\circ = \cos 18^\circ]$$

$$\Rightarrow \text{LHS} = \left\{ \frac{\sqrt{10 + 2\sqrt{5}}}{4} \right\}^2 - \left(\frac{\sqrt{3}}{2} \right)^2 = \frac{10 + 2\sqrt{5}}{16} - \frac{3}{4} = \frac{2\sqrt{5} - 2}{16} = \frac{\sqrt{5} - 1}{8} = \text{RHS}$$

(ii) We have,

$$\text{LHS} = \cos^2 48^\circ - \sin^2 12^\circ$$

$$\Rightarrow \text{LHS} = \cos (48^\circ + 12^\circ) \cos (48^\circ - 12^\circ) \quad [\because \cos^2 A - \sin^2 B = \cos (A + B) \cos (A - B)]$$

$$\Rightarrow \text{LHS} = \cos 60^\circ \cos 36^\circ = \frac{1}{2} \times \frac{\sqrt{5} + 1}{4} = \frac{\sqrt{5} + 1}{8} = \text{RHS}$$

(iii) We have,

$$\text{LHS} = \sin \frac{\pi}{10} + \sin \frac{13\pi}{10}$$

$$\Rightarrow \text{LHS} = \sin 18^\circ + \sin 234^\circ = \sin 18^\circ + \sin (270^\circ - 36^\circ)$$

$$\Rightarrow \text{LHS} = \sin 18^\circ - \cos 36^\circ = \frac{\sqrt{5} - 1}{4} - \frac{\sqrt{5} + 1}{4} = -\frac{1}{2}$$

(iv) We have,

$$\Rightarrow \text{LHS} = \sin \frac{\pi}{10} \sin \frac{13\pi}{10}$$

$$\Rightarrow \text{LHS} = \sin 18^\circ \sin 234^\circ = -\sin 18^\circ \cos 36^\circ = -\frac{\sqrt{5}-1}{4} \times \frac{\sqrt{5}+1}{4} = -\left(\frac{5-1}{16}\right) = -\frac{1}{4} = \text{RHS}$$

(v) We have,

$$\text{LHS} = \sin^2 24^\circ - \sin^2 6^\circ$$

$$\Rightarrow \text{LHS} = \sin (24^\circ + 6^\circ) \sin (24^\circ - 6^\circ) \quad [\because \sin (A+B) \sin (A-B) = \sin^2 A - \sin^2 B]$$

$$\Rightarrow \text{LHS} = \sin 30^\circ \sin 18^\circ = \frac{1}{2} \times \frac{\sqrt{5}-1}{4} = \frac{\sqrt{5}-1}{8} = \text{RHS}$$

EXAMPLE 2 Prove that: $\sin \frac{\pi}{5} \sin \frac{2\pi}{5} \sin \frac{3\pi}{5} \sin \frac{4\pi}{5} = \frac{5}{16}$

SOLUTION If $A+B=\pi$, then $A=\pi-B \Rightarrow \sin A = \sin (\pi-B) \Rightarrow \sin A = \sin B$

$$\therefore \frac{\pi}{5} + \frac{4\pi}{5} = \pi \Rightarrow \sin \frac{\pi}{5} = \sin \frac{4\pi}{5} \text{ and } \frac{2\pi}{5} + \frac{3\pi}{5} = \pi \Rightarrow \sin \frac{2\pi}{5} = \sin \frac{3\pi}{5}$$

Using these values, we obtain

$$\text{LHS} = \sin \frac{\pi}{5} \sin \frac{2\pi}{5} \sin \frac{3\pi}{5} \sin \frac{4\pi}{5}$$

$$\Rightarrow \text{LHS} = \sin \frac{\pi}{5} \sin \frac{2\pi}{5} \sin \frac{2\pi}{5} \sin \frac{\pi}{5}$$

$$\Rightarrow \text{LHS} = \left(\sin \frac{\pi}{5} \sin \frac{2\pi}{5} \right)^2 = (\sin 36^\circ \sin 72^\circ)^2 = (\sin 36^\circ \cos 18^\circ)^2$$

$$\Rightarrow \text{LHS} = \left\{ \frac{\sqrt{10-2\sqrt{5}}}{4} \times \frac{\sqrt{10+2\sqrt{5}}}{4} \right\}^2 = \frac{10-2\sqrt{5}}{16} \times \frac{10+2\sqrt{5}}{16} = \frac{100-20}{256} = \frac{80}{256} = \frac{5}{16} = \text{RHS}$$

EXAMPLE 3 Prove that: $16 \cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{8\pi}{15} \cos \frac{14\pi}{15} = 1$

SOLUTION We have,

$$\text{LHS} = 16 \cos 24^\circ \cos 48^\circ \cos 96^\circ \cos 168^\circ$$

$$\Rightarrow \text{LHS} = 4(2 \cos 24^\circ \cos 96^\circ)(2 \cos 48^\circ \cos 168^\circ)$$

$$\Rightarrow \text{LHS} = 4(\cos 120^\circ + \cos 72^\circ)(\cos 216^\circ + \cos 120^\circ)$$

$$\Rightarrow \text{LHS} = 4(-\sin 30^\circ + \sin 18^\circ)(-\cos 36^\circ - \sin 30^\circ)$$

$$\Rightarrow \text{LHS} = 4\left(-\frac{1}{2} + \frac{\sqrt{5}-1}{4}\right)\left(-\frac{\sqrt{5}+1}{4} - \frac{1}{2}\right)$$

$$\Rightarrow \text{LHS} = 4\left(\frac{\sqrt{5}-3}{4}\right)\left(\frac{-\sqrt{5}-3}{4}\right) = 4\left(\frac{3-\sqrt{5}}{4}\right)\left(\frac{3+\sqrt{5}}{4}\right) = \left(\frac{9-5}{4}\right) = 1 = \text{RHS}$$

EXAMPLE 4 Prove that: $\sin 12^\circ \sin 48^\circ \sin 54^\circ = \frac{1}{8}$.

SOLUTION We have,

$$\text{LHS} = \frac{1}{2}(2 \sin 48^\circ \sin 12^\circ) \sin 54^\circ$$

$$\Rightarrow \text{LHS} = \frac{1}{2}(\cos 36^\circ - \cos 60^\circ) \cos 36^\circ \quad [\because 2 \sin A \sin B = \cos (A-B) - \cos (A+B)]$$

$$\Rightarrow \text{LHS} = \frac{1}{2}\left(\frac{\sqrt{5}+1}{4} - \frac{1}{2}\right)\left(\frac{\sqrt{5}+1}{4}\right) = \frac{1}{2}\left(\frac{\sqrt{5}-1}{4}\right)\left(\frac{\sqrt{5}+1}{4}\right) = \frac{1}{8} = \text{RHS}$$

EXAMPLE 5 Prove that: $\left(1 + \cos \frac{\pi}{10}\right)\left(1 + \cos \frac{3\pi}{10}\right)\left(1 + \cos \frac{7\pi}{10}\right)\left(1 + \cos \frac{9\pi}{10}\right) = \frac{1}{16}$.

SOLUTION If $A + B = \pi$, then $\cos A = \cos(\pi - B) = -\cos B$

$$\therefore \frac{\pi}{10} + \frac{9\pi}{10} = \pi \Rightarrow \cos \frac{9\pi}{10} = -\cos \frac{\pi}{10} \text{ and, } \frac{3\pi}{10} + \frac{7\pi}{10} = \pi \Rightarrow \cos \frac{7\pi}{10} = -\cos \frac{3\pi}{10}$$

Using these values, we obtain

$$\begin{aligned} \text{LHS} &= \left(1 + \cos \frac{\pi}{10}\right)\left(1 + \cos \frac{3\pi}{10}\right)\left(1 - \cos \frac{3\pi}{10}\right)\left(1 - \cos \frac{\pi}{10}\right) \\ \Rightarrow \text{LHS} &= \left(1 - \cos^2 \frac{\pi}{10}\right)\left(1 - \cos^2 \frac{3\pi}{10}\right) \\ \Rightarrow \text{LHS} &= (1 - \cos^2 18^\circ)(1 - \cos^2 54^\circ) = \sin^2 18^\circ \sin^2 54^\circ = \sin^2 18^\circ \cos^2 36^\circ \\ \Rightarrow \text{LHS} &= (\sin 18^\circ \cos 36^\circ)^2 = \left(\frac{\sqrt{5}-1}{4} \times \frac{\sqrt{5}+1}{4}\right)^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16} = \text{RHS} \end{aligned}$$

EXAMPLE 6 Prove that: $\tan 6^\circ \tan 42^\circ \tan 66^\circ \tan 78^\circ = 1$.

SOLUTION We have,

$$\begin{aligned} \text{LHS} &= \frac{\sin 6^\circ \sin 42^\circ \sin 66^\circ \sin 78^\circ}{\cos 6^\circ \cos 42^\circ \cos 66^\circ \cos 78^\circ} = \frac{(2 \sin 66^\circ \sin 6^\circ)(2 \sin 78^\circ \sin 42^\circ)}{(2 \cos 66^\circ \cos 6^\circ)(2 \cos 78^\circ \cos 42^\circ)} \\ \Rightarrow \text{LHS} &= \frac{(\cos 60^\circ - \cos 72^\circ)(\cos 36^\circ - \cos 120^\circ)}{(\cos 60^\circ + \cos 72^\circ)(\cos 36^\circ + \cos 120^\circ)} \\ \Rightarrow \text{LHS} &= \frac{(\cos 60^\circ - \sin 18^\circ)(\cos 36^\circ + \sin 30^\circ)}{(\cos 60^\circ + \sin 18^\circ)(\cos 36^\circ - \sin 30^\circ)} \\ \Rightarrow \text{LHS} &= \frac{\left(\frac{1}{2} - \frac{\sqrt{5}-1}{4}\right)\left(\frac{\sqrt{5}+1}{4} + \frac{1}{2}\right)}{\left(\frac{1}{2} + \frac{\sqrt{5}-1}{4}\right)\left(\frac{\sqrt{5}-1}{4} - \frac{1}{2}\right)} = \frac{(3-\sqrt{5})(3+\sqrt{5})}{(\sqrt{5}+1)(\sqrt{5}-1)} = \frac{9-5}{5-1} = 1 = \text{RHS} \end{aligned}$$

LEVEL-2

EXAMPLE 7 Prove that : $4 \sin 27^\circ = \sqrt{5+\sqrt{5}} - \sqrt{3-\sqrt{5}}$.

SOLUTION We have,

$$\begin{aligned} 16 \sin^2 27^\circ &= 8(2 \sin^2 27^\circ) = 8(1 - \cos 54^\circ) = 8(1 - \sin 36^\circ) \\ &= 8 \left\{ 1 - \frac{\sqrt{10-2\sqrt{5}}}{4} \right\} = 2 \left\{ 4 - \sqrt{10-2\sqrt{5}} \right\} = 8 - 2\sqrt{10-2\sqrt{5}} \\ &= (5+\sqrt{5}) + (3-\sqrt{5}) - 2\sqrt{(5+\sqrt{5})(3-\sqrt{5})} \\ &= \left\{ \sqrt{5+\sqrt{5}} \right\}^2 + \left\{ \sqrt{3-\sqrt{5}} \right\}^2 - 2\sqrt{(5+\sqrt{5})(3-\sqrt{5})} \\ &= \left\{ \sqrt{5+\sqrt{5}} - \sqrt{3-\sqrt{5}} \right\}^2 \end{aligned}$$

Taking square roots of both sides, we obtain

$$\therefore 4 \sin 27^\circ = \sqrt{5+\sqrt{5}} - \sqrt{3-\sqrt{5}}$$

[$\because \sin 27^\circ$ is positive]

EXAMPLE 8 Find the value of $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$.

SOLUTION $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$

$$= (\tan 9^\circ + \tan 81^\circ) - (\tan 27^\circ + \tan 63^\circ)$$

$$= (\tan 9^\circ + \cot 9^\circ) - (\tan 27^\circ + \cot 27^\circ)$$

$$\left[\begin{array}{l} \because \tan 81^\circ = \tan (90^\circ - 9^\circ) = \cot 9^\circ \\ \tan 63^\circ = \tan (90^\circ - 27^\circ) = \cot 27^\circ \end{array} \right]$$

$$= \frac{1}{\sin 9^\circ \cos 9^\circ} - \frac{1}{\sin 27^\circ \cos 27^\circ}$$

$$\left[\because \tan \theta + \cot \theta = \frac{1}{\sin \theta \cos \theta} \right]$$

$$= \frac{2}{\sin 18^\circ} - \frac{2}{\sin 54^\circ} = \frac{2}{\sin 18^\circ} - \frac{2}{\cos 36^\circ} = \frac{8}{\sqrt{5}-1} - \frac{8}{\sqrt{5}+1} = \frac{8 \times 2}{5-1} = 4$$

EXERCISE 9.3

LEVEL-1

Prove that:

$$1. \sin^2 72^\circ - \sin^2 60^\circ = \frac{\sqrt{5}-1}{8}$$

$$2. \sin^2 24^\circ - \sin^2 6^\circ = \frac{\sqrt{5}-1}{8}$$

$$3. \sin^2 42^\circ - \cos^2 78^\circ = \frac{\sqrt{5}+1}{8}$$

$$4. \cos 78^\circ \cos 42^\circ \cos 36^\circ = \frac{1}{8}$$

$$5. \cos \frac{\pi}{15} \cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{7\pi}{15} = \frac{1}{16}$$

$$6. \cos \frac{\pi}{15} \cos \frac{2\pi}{15} \cos \frac{3\pi}{15} \cos \frac{4\pi}{15} \cos \frac{5\pi}{15} \cos \frac{6\pi}{15} \cos \frac{7\pi}{15} = \frac{1}{128}$$

$$7. \cos 6^\circ \cos 42^\circ \cos 66^\circ \cos 78^\circ = \frac{1}{16}$$

$$8. \sin 6^\circ \sin 42^\circ \sin 66^\circ \sin 78^\circ = \frac{1}{16}$$

$$9. \cos 36^\circ \cos 42^\circ \cos 60^\circ \cos 78^\circ = \frac{1}{16}$$

$$10. \sin 36^\circ \sin 72^\circ \sin 108^\circ \sin 144^\circ = \frac{5}{16}$$

VERY SHORT ANSWER QUESTIONS (VSAQs)

Answer each of the following questions in one word or one sentence or as per exact requirement of the question:

$$1. \text{ If } \cos 4x = 1 + k \sin^2 x \cos^2 x, \text{ then write the value of } k.$$

$$2. \text{ If } \tan \frac{x}{2} = \frac{m}{n}, \text{ then write the value of } m \sin x + n \cos x.$$

$$3. \text{ If } \frac{\pi}{2} < \theta < \frac{3\pi}{2}, \text{ then write the value of } \sqrt{\frac{1 + \cos 2\theta}{2}}.$$

$$4. \text{ If } \frac{\pi}{2} < \theta < \pi, \text{ then write the value of } \sqrt{2 + \sqrt{2 + 2 \cos 2\theta}} \text{ in the simplest form.}$$

$$5. \text{ If } \frac{\pi}{2} < \theta < \pi, \text{ then write the value of } \sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}}.$$

6. If $\pi < \theta < \frac{3\pi}{2}$, then write the value of $\sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}}$.
7. In a right angled triangle ABC , write the value of $\sin^2 A + \sin^2 B + \sin^2 C$.
8. Write the value of $\cos^2 76^\circ + \cos^2 16^\circ - \cos 76^\circ \cos 16^\circ$.
9. If $\frac{\pi}{4} < \theta < \frac{\pi}{2}$, then write the value of $\sqrt{1 - \sin 2\theta}$.
10. Write the value of $\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7}$.
11. If $\tan A = \frac{1 - \cos B}{\sin B}$, then find the value of $\tan 2A$.
12. If $\sin x + \cos x = a$, find the value of $\sin^6 x + \cos^6 x$.
13. If $\sin x + \cos x = a$, find the value of $|\sin x - \cos x|$.

ANSWERS

1. -8 2. n 3. $-\cos \theta$ 4. $2 \sin \frac{\theta}{2}$ 5. $-\tan \theta$ 6. $\tan \theta$ 7. 2
8. $\frac{3}{4}$ 9. $\sin \theta - \cos \theta$ 10. $-\frac{1}{8}$ 11. $\tan B$ 12. $\frac{1}{4} \{4 - 3(a^2 - 1)^2\}$ 13. $\sqrt{2 - a^2}$.

MULTIPLE CHOICE QUESTIONS (MCQs)

Mark the correct alternative in each of the following:

1. $8 \sin \frac{x}{8} \cos \frac{x}{2} \cos \frac{x}{4} \cos \frac{x}{8}$ is equal to
 (a) $8 \cos x$ (b) $\cos x$ (c) $8 \sin x$ (d) $\sin x$
2. $\frac{\sec 8A - 1}{\sec 4A - 1}$ is equal to
 (a) $\frac{\tan 2A}{\tan 8A}$ (b) $\frac{\tan 8A}{\tan 2A}$ (c) $\frac{\cot 8A}{\cot 2A}$ (d) none of these
3. The value of $\cos \frac{\pi}{65} \cos \frac{2\pi}{65} \cos \frac{4\pi}{65} \cos \frac{8\pi}{65} \cos \frac{16\pi}{65} \cos \frac{32\pi}{65}$ is
 (a) $\frac{1}{8}$ (b) $\frac{1}{16}$ (c) $\frac{1}{32}$ (d) none of these
4. If $\cos 2x + 2 \cos x = 1$ then, $(2 - \cos^2 x) \sin^2 x$ is equal to
 (a) 1 (b) -1 (c) $-\sqrt{5}$ (d) $\sqrt{5}$
5. For all real values of x , $\cot x - 2 \cot 2x$ is equal to
 (a) $\tan 2x$ (b) $\tan x$ (c) $-\cot 3x$ (d) none of these
6. The value of $2 \tan \frac{\pi}{10} + 3 \sec \frac{\pi}{10} - 4 \cos \frac{\pi}{10}$ is
 (a) 0 (b) $\sqrt{5}$ (c) 1 (d) none of these
7. If in a ΔABC , $\tan A + \tan B + \tan C = 0$, then $\cot A \cot B \cot C =$
 (a) 6 (b) 1 (c) $\frac{1}{6}$ (d) none of these
8. If $\cos \theta = \frac{1}{2} \left(a + \frac{1}{a} \right)$, and $\cos 3\theta = \lambda \left(a^3 + \frac{1}{a^3} \right)$, then $\lambda =$

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) 1 (d) none of these
9. If $2 \tan \alpha = 3 \tan \beta$, then $\tan(\alpha - \beta) =$
 (a) $\frac{\sin 2\beta}{5 - \cos 2\beta}$ (b) $\frac{\cos 2\beta}{5 - \cos 2\beta}$ (c) $\frac{\sin 2\beta}{5 + \cos 2\beta}$ (d) none of these
10. If $\tan \alpha = \frac{1 - \cos \beta}{\sin \beta}$, then
 (a) $\tan 3\alpha = \tan 2\beta$ (b) $\tan 2\alpha = \tan \beta$
 (c) $\tan 2\beta = \tan \alpha$ (d) none of these
11. If $\sin \alpha + \sin \beta = a$ and $\cos \alpha - \cos \beta = b$, then $\tan \frac{\alpha - \beta}{2} =$
 (a) $-\frac{a}{b}$ (b) $-\frac{b}{a}$ (c) $\sqrt{a^2 + b^2}$ (d) none of these
12. The value of $\left(\cot \frac{x}{2} - \tan \frac{x}{2}\right)^2 (1 - 2 \tan x \cot 2x)$ is
 (a) 1 (b) 2 (c) 3 (d) 4
13. The value of $\tan \theta \sin\left(\frac{\pi}{2} + \theta\right) \cos\left(\frac{\pi}{2} - \theta\right)$ is
 (a) 1 (b) -1 (c) $\frac{1}{2} \sin 2\theta$ (d) none of these
14. The value of $\sin^2\left(\frac{\pi}{18}\right) + \sin^2\left(\frac{\pi}{9}\right) + \sin^2\left(\frac{7\pi}{18}\right) + \sin^2\left(\frac{4\pi}{9}\right)$ is
 (a) 1 (b) 2 (c) 4 (d) none of these
15. If $5 \sin \alpha = 3 \sin(\alpha + 2\beta) \neq 0$, then $\tan(\alpha + \beta)$ is equal to
 (a) $2 \tan \beta$ (b) $3 \tan \beta$ (c) $4 \tan \beta$ (d) $6 \tan \beta$
16. The value of $2 \cos \theta - \cos 3\theta - \cos 5\theta - 16 \cos^3 \theta \sin^2 \theta$ is
 (a) 2 (b) 1 (c) 0 (d) -1
17. If $A = 2 \sin^2 \theta - \cos 2\theta$, then A lies in the interval
 (a) $[-1, 3]$ (b) $[1, 2]$ (c) $[-2, 4]$ (d) none of these
18. The value of $\frac{\cos 3\theta}{2 \cos 2\theta - 1}$ is equal to
 (a) $\cos \theta$ (b) $\sin \theta$ (c) $\tan \theta$ (d) none of these
19. If $\tan(\pi/4 + \theta) + \tan(\pi/4 - \theta) = \lambda \sec 2\theta$, then
 (a) 3 (b) 4 (c) 1 (d) 2
20. The value of $\cos^2\left(\frac{\pi}{6} + \theta\right) - \sin^2\left(\frac{\pi}{6} - \theta\right)$ is
 (a) $\frac{1}{2} \cos 2\theta$ (b) 0 (c) $-\frac{1}{2} \cos 2\theta$ (d) $\frac{1}{2}$
21. $\frac{\sin 3\theta}{1 + 2 \cos 2\theta}$ is equal to
 (a) $\cos \theta$ (b) $\sin \theta$ (c) $-\cos \theta$ (d) $\sin \theta$
22. The value of $2 \sin^2 B + 4 \cos(A + B) \sin A \sin B + \cos 2(A + B)$ is
 (a) 0 (b) $\cos 3A$ (c) $\cos 2A$ (d) none of these

CHAPTER 10

SINE AND COSINE FORMULAE AND THEIR APPLICATIONS

10.1 INTRODUCTION

In any triangle the three sides and the three angles are generally called the elements of the triangle. A triangle which does not contain a right angle is called an *oblique triangle*.

In any triangle ABC , the measures of the angles $\angle BAC$, $\angle CBA$ and $\angle ACB$ are denoted by the letters A , B and C respectively. The sides BC , CA and AB opposite to the angles A , B and C respectively are denoted by a , b and c . These six elements of a triangle are not independent and are connected by the relations: (i) $A + B + C = \pi$ (ii) $a + b > c$; $b + c > a$; $c + a > b$. In addition to these relations, the elements of a triangle are connected by some trigonometric relations. We intend to discuss those relations in the sections to follow of this chapter.

10.2 THE LAW OF SINES OR SINE RULE

THEOREM *The sides of a triangle are proportional to the sines of the angles opposite to them i.e. in a ΔABC ,*

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

PROOF The following cases arise:

CASE I *When ΔABC is an acute angled triangle:*

Draw AD perpendicular from A to the opposite side BC meeting it in the point D .

In the triangle ABD , we have

$$\sin B = \frac{AD}{AB} \Rightarrow \sin B = \frac{AD}{c} \Rightarrow AD = c \sin B \quad \dots(i)$$

In the triangle ACD , we have

$$\sin C = \frac{AD}{AC} \Rightarrow \sin C = \frac{AD}{b} \Rightarrow AD = b \sin C \quad \dots(ii)$$

From (i) and (ii), we get

$$c \sin B = b \sin C \Rightarrow \frac{b}{\sin B} = \frac{c}{\sin C}$$

In a similar manner, by drawing a perpendicular from B on AC , we obtain

$$\frac{a}{\sin A} = \frac{c}{\sin C}$$

Hence,
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

CASE II *When ΔABC is an obtuse angled triangle.*

Draw AD perpendicular from A on CB produced meeting it in D .

In ΔADB , we have

$$\sin \angle ABD = \frac{AD}{AB} \Rightarrow \sin (180 - B) = \frac{AD}{c} \Rightarrow \sin B = \frac{AD}{c} \Rightarrow AD = c \sin B \quad \dots(i)$$

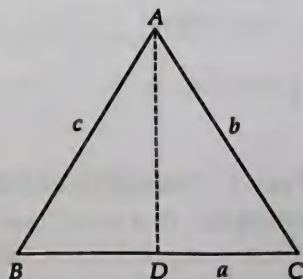


Fig. 10.1

In $\triangle ACD$, we have

$$\sin C = \frac{AD}{AC} \Rightarrow \sin C = \frac{AD}{b} \Rightarrow AD = b \sin C \quad \dots(ii)$$

From (i) and (ii), we obtain

$$c \sin B = b \sin C \Rightarrow \frac{b}{\sin B} = \frac{c}{\sin C}$$

Similarly, by drawing perpendicular from B on AC , we obtain

$$\frac{a}{\sin A} = \frac{c}{\sin C}$$

$$\text{Hence, } \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

CASE III When $\triangle ABC$ is a right angled triangle:

In $\triangle ABC$, we have

$$\sin C = \sin \frac{\pi}{2} = 1, \sin A = \frac{BC}{AB} = \frac{a}{c} \text{ and, } \sin B = \frac{AC}{AB} = \frac{b}{c}$$

$$\therefore \frac{a}{\sin A} = \frac{b}{\sin B} = c$$

$$\Rightarrow \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{1}$$

$$\Rightarrow \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \quad \left[\because \sin C = \sin \frac{\pi}{2} = 1 \right]$$

Hence, in all the cases, we obtain

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

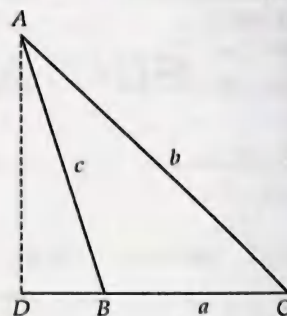


Fig. 10.2

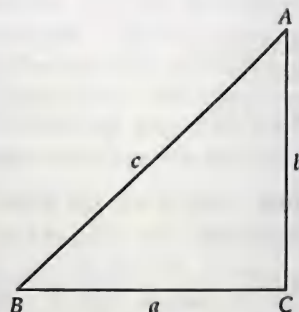


Fig. 10.3

Q.E.D.

REMARK 1 The above rule may also be expressed as $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

REMARK 2 The sine rule is a very useful tool to express sides of a triangle in terms of the sines of angles and vice-versa in the following manner.

Let $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$ (say) Then, $a = k \sin A$, $b = k \sin B$, $c = k \sin C$.

Similarly,

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} = \lambda \text{ (say)} \Rightarrow \sin A = \lambda a, \sin B = \lambda b \text{ and } \sin C = \lambda c.$$

ILLUSTRATIVE EXAMPLES

LEVEL-1

Type I PROBLEMS BASED ON SINE RULE

RESULTS In a $\triangle ABC$, we have

$$\sin (B + C) = \sin A, \sin (C + A) = \sin B, \sin (A + B) = \sin C$$

$$\cos (B + C) = -\cos A, \cos (C + A) = -\cos B, \cos (A + B) = -\cos C$$

$$\tan (B + C) = -\tan A, \tan (C + A) = -\tan B, \tan (A + B) = -\tan C$$

EXAMPLE 1 In a $\triangle ABC$, if $a = 2$, $b = 3$ and $\sin A = \frac{2}{3}$, find $\angle B$.

SOLUTION We have,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\Rightarrow \frac{2}{(2/3)} = \frac{3}{\sin B} \Rightarrow 3 = \frac{3}{\sin B} \Rightarrow \sin B = 1 \Rightarrow \angle B = 90^\circ$$

EXAMPLE 2 If in any triangle the angles be to one another as $1 : 2 : 3$, prove that the corresponding sides are $1 : \sqrt{3} : 2$.

SOLUTION Let the measures of the angles be $x, 2x$ and $3x$. Then,

$$x + 2x + 3x = 180^\circ \Rightarrow 6x = 180^\circ \Rightarrow x = 30^\circ$$

So, the angles are $30^\circ, 60^\circ$ and 90°

Let a, b, c be the lengths of the sides apposite to these angles. Then,

$$\frac{a}{\sin 30^\circ} = \frac{b}{\sin 60^\circ} = \frac{c}{\sin 90^\circ}$$

$$\Rightarrow a : b : c = \sin 30^\circ : \sin 60^\circ : \sin 90^\circ$$

$$\Rightarrow a : b : c = \frac{1}{2} : \frac{\sqrt{3}}{2} : 1 \Rightarrow a : b : c = 1 : \sqrt{3} : 2$$

EXAMPLE 3 The angles of a triangle ABC are in A.P. and it is being given that $b : c = \sqrt{3} : \sqrt{2}$, find $\angle A$.

SOLUTION It is given that the angles $\angle A, \angle B, \angle C$ are in A.P.

$$\therefore 2\angle B = \angle A + \angle C \Rightarrow 3\angle B = \angle A + \angle B + \angle C \Rightarrow 3\angle B = 180^\circ \Rightarrow \angle B = 60^\circ$$

Now, $\frac{b}{\sin B} = \frac{c}{\sin C}$

$$\Rightarrow \frac{b}{c} = \frac{\sin B}{\sin C}$$

$$\Rightarrow \frac{\sqrt{3}}{\sqrt{2}} = \frac{\sin 60^\circ}{\sin C}$$

$$[\because b : c = \sqrt{3} : \sqrt{2}]$$

$$\Rightarrow \frac{\sqrt{3}}{\sqrt{2}} = \frac{\sqrt{3}/2}{\sin C}$$

$$\Rightarrow \sin C = \frac{1}{\sqrt{2}} \Rightarrow \angle C = 45^\circ$$

$$\therefore \angle A = 180^\circ - (\angle B + \angle C) = 180^\circ - (60^\circ + 45^\circ) = 75^\circ$$

EXAMPLE 4 In any triangle ABC , prove that:

$$(i) \frac{\sin(B-C)}{\sin(B+C)} = \frac{b^2 - c^2}{a^2}$$

[NCERT]

$$(ii) a \sin(B-C) + b \sin(C-A) + c \sin(A-B) = 0$$

[NCERT]

$$(iii) a^3 \sin(B-C) + b^3 \sin(C-A) + c^3 \sin(A-B) = 0$$

SOLUTION Let $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k$. Then,

$$a = k \sin A, b = k \sin B \text{ and } c = k \sin C.$$

...(i)

$$(i) \text{ RHS} = \frac{b^2 - c^2}{a^2} = \frac{k^2 \sin^2 B - k^2 \sin^2 C}{k^2 \sin^2 A}$$

[Using (i)]

$$\Rightarrow \text{RHS} = \frac{\sin^2 B - \sin^2 C}{\sin^2 A} = \frac{\sin(B+C) \sin(B-C)}{\sin^2 A}$$

$$\Rightarrow \text{RHS} = \frac{\sin(\pi - A) \sin(B-C)}{\sin^2 A}$$

$$[\because A+B+C = \pi \Rightarrow B+C = \pi - A]$$